

ENVIRONMENTAL RESEARCH
LETTERS

TOPICAL REVIEW

OPEN ACCESS

RECEIVED

28 February 2026

REVISED

28 April 2026

ACCEPTED FOR PUBLICATION

22 June 2026

PUBLISHED

8 July 2026

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Wildfire as urban risk: global synthesis of compound hazards,
cascade pathways, and research-exposure-vulnerability mismatchXinyi Zhang¹ and Lu Liang^{1*}

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E-mail: lianglu@berkeley.edu**Keywords:** fire weather, IPCC, infrastructure, multi-hazard interaction, wildland-urban interface, LLM, climate extremeSupplementary material for this article is available [online](#)

Abstract

As cities expand into fire-prone landscapes and climate change intensifies both wildfire activity and the associated climate extremes, wildfire is increasingly an urban risk. However, wildfire-linked risks have been inconsistently defined across disciplines, limiting cross-study comparison and policy translation. Compound climate conditions increasingly drive fire occurrence, but how specific combinations shape risk across fire regime types remains poorly resolved. Urban and wildland-urban interface (WUI) areas face rapidly growing fire exposure, while research concentrates on wildland settings and low-deprivation areas, leaving the most fire-exposed and socially vulnerable urban populations underrepresented in the literature. Here we review 3625 studies on wildfire-linked risks (Web of Science, 2000–2025) alongside global spatial risk analysis of fire occurrence, compound climate conditions, land use setting, and social vulnerability. We develop an interaction taxonomy using decision-tree logic that not only resolves terminological inconsistencies but also enables alignment with distinct management phases. Compound climate conditions, particularly co-occurring heat and drought, are the strongest predictors of wildfire occurrence globally, but their effects vary sharply across fire regime types. A triple-mismatch framework reveals that fire burden, research neglect, and social vulnerability converge most severely in the Global South. WUI and urban areas account for over 33% of global fire occurrence but receive only 4% of research attention. Urban fire risk depends on compound climate extremes rather than any single weather driver. Single climate extremes produce negligible risk elevation, yet compound conditions produce stronger risk amplification in urban areas than in any other setting. Smoke transport, debris flows, and infrastructure cascades emerge as the dominant propagation pathways through urban systems. Finally, based on these findings, we propose priority actions in compound hazard assessment, WUI-focused monitoring, equity-aware research, and phase-specific management.

1. Introduction: wildfire as urban risk

In January 2025, the Palisades and Eaton fires burned through densely populated neighborhoods in Los Angeles during a multi-month drought and powerful Santa Ana winds, destroying over 16 000 structures and causing tens of billions of dollars in direct losses [1]. During Australia's 2019–20 Black Summer, multi-year drought and record-breaking heat fueled bushfires that burned over 10 million hectares and blanketed Sydney and Melbourne in

hazardous smoke for weeks, exposing millions of urban residents to sustained poor air quality [2]. In 2018, intense rainfall on the recently burned, drought-primed Thomas Fire watershed produced a post-fire debris flow at Montecito, California, that caused more casualties than the preceding fire itself [3, 4]. These events share a common thread: each illustrates how compound climate drivers produce wildfire-linked risks that reach urban populations. Although wildfires rarely burn through dense urban cores, the compound drivers intensified the

fires themselves, while the linked risks that followed (smoke transport, debris flows, and infrastructure failures) propagated far beyond the burn perimeter into the urban systems we examine throughout this review.

Wildfire is no longer a remote wildland phenomenon. As urban development expands into fire-prone landscapes and climate change intensifies fire weather, metropolitan regions face growing exposure. This exposure takes several forms: direct burning at the wildland-urban interface (WUI), smoke reaching cities hundreds of kilometers from active fires, and post-fire geomorphic hazards in communities below burn areas [5–7]. Intergovernmental Panel on Climate Change (IPCC) assessment reports characterize risk as arising from the intersection of hazard, exposure, and vulnerability [8, 9]. In urban fire contexts, these three components intensify simultaneously. Compound interactions between drought, heat, wind, and post-fire rainfall challenge conventional single-hazard frameworks and require integrated approaches to urban risk assessment [10–12].

Substantial research has advanced understanding of fire as a climate-sensitive Earth-system process [13, 14] and of compound weather and climate events more broadly [11, 12, 15, 16]. Successive IPCC assessment reports have given increasing attention to compound-event and cascading-risk frameworks [8, 9, 17]. Recent syntheses document the rapid growth of compound-extremes research spanning floods, droughts, heatwaves, and wildfires [18]. However, most existing reviews focus on conceptual frameworks or particular hazard combinations at regional to global scales. Three gaps limit the translation of this growing evidence base into urban systems.

First, definitions and interaction types are inconsistent across disciplines. Atmospheric scientists describe drought-fire relationships as ‘preconditioned’ events. Geomorphologists classify fire-to-debris-flow sequences as ‘triggered’ cascades. Emergency managers speak of ‘compound’ disasters when smoke and heat overlap. These terminological differences are not merely semantic. They reflect distinct assumptions about causality, timing, and intervention point and limit comparability across studies and hinder communication between researchers and practitioners [12, 19, 20]. Without a shared classification framework, synthesizing evidence across disciplines remains difficult, and policy guidance drawn from one framing may not transfer to another.

Second, the evidence base remains skewed toward wildland settings. The WUI is the fastest-growing land use type in the United States [5], and global WUI extent is expanding rapidly [6]. Yet wildfire research overwhelmingly studies fire behavior, ecology, and climate drivers in non-urban landscapes. Urban and WUI environments concentrate populations, critical infrastructure, and interconnected services in ways that reshape how wildfire-linked risks propagate. The

same post-fire rainfall that produces a debris flow in a remote watershed can destroy water treatment plants, sever evacuation routes, and trigger cascading infrastructure failures when it occurs near a city [17, 20]. These distinctions receive limited systematic attention.

Third, social vulnerability is often treated as an afterthought or omitted entirely. In California, homes in disadvantaged communities face 29% higher probability of wildfire destruction [21]. Compound events amplify these disparities because adapting to simultaneous or sequential hazards requires resources, mobility, and institutional support that are unequally distributed [22]. Globally, the populations most exposed to wildfire-linked risks often live in regions with the least research coverage and the weakest adaptive capacity. Understanding who is at risk, and why, is essential for designing interventions that do not reproduce existing inequities.

This review addresses these three gaps by combining literature synthesis with global spatial analysis. We systematically reviewed 3625 peer-reviewed studies (2000–2025), and separately constructed a global spatial analysis of fire occurrence, compound climate conditions, land-use setting, and social vulnerability. To classify this volume of literature consistently, we used a large language model (LLM) assisted workflow validated against manual review, which enables consistent, transparent coding across the full corpus. Satellite-derived hazard and exposure layers then allow direct comparison between where research concentrates and where risks actually occur. Our analysis is structured around three questions:

- RQ1: How have wildfire-linked hazard definitions and interaction types evolved, and can they be organized into a consistent classification framework?
- RQ2: How do compound climate drivers shape wildfire risk across fire regimes and along the wildland-to-urban gradient, and where does this risk distribution converge with or diverge from research attention and social vulnerability?
- RQ3: What strategies and pathways can bridge the gaps between current evidence and the needs of urban wildfire-linked risk management?

2. Objectives and integrated analytical framework

The three gaps identified in introduction cannot be closed by a literature synthesis or a spatial analysis in isolation. Resolving inconsistent terminology calls for a classification grounded in the language researchers actually use. Locating where fire, research, and vulnerability converge, or fail to converge, requires empirical measurement beyond what

the literature alone can provide. This review therefore adopts an integrated analytical framework in which the literature synthesis and the spatial analysis contribute to a single account of risk.

RQ1 asks how inconsistent terminology across disciplines can be organized into a usable classification. We address it through a hazard interaction taxonomy derived from the literature corpus, extended with a causal chain branch that captures urban cascades. The taxonomy provides a common vocabulary for both published studies and the climate-fire relationships measured in the spatial analysis.

RQ2 asks where wildfire-linked risks concentrate globally, and how compound climate conditions shape fire occurrence across regions, fire regimes, and urban settings. Two components work together to answer it. A compound hazard layer, built from global earth observation and reanalysis data, quantifies where specific climate driver combinations elevate fire occurrence and how they vary along the wildland to urban gradient. A Research-Risk Mismatch framework then overlays geolocated study attention and the Global Gridded Relative Deprivation Index on that hazard layer, placing physical risk, research attention, and social vulnerability in direct spatial comparison.

RQ3 asks what priorities for urban wildfire risk management follow from this combined evidence. We answer it by drawing the taxonomy and the mismatch findings together. The taxonomy links each interaction type to a distinct management phase, from prevention through recovery. The mismatch findings locate where those phases are most urgent and where the evidence base is weakest. Read jointly, they identify targeted priorities for research, monitoring, policy, and planning.

The three components align with the IPCC risk concept of hazard, exposure, and vulnerability. The taxonomy clarifies what kinds of hazard interactions are being compared. The compound hazard analysis identifies where those hazards concentrate globally. The wildland urban interface classification and the deprivation overlay populate the exposure and vulnerability dimensions respectively. The literature synthesis and the spatial analysis are not parallel studies but complementary contributions to a single argument. The synthesis provides the classification system, the distribution of research attention, and the published record of reported cascades. The spatial analysis provides an independent measurement of where compound hazards actually occur and which populations they reach.

Section 4 establishes the taxonomy. Section 5 applies the compound hazard and mismatch layers at the global and regional scale. Section 6 extends them across the wildland to urban gradient. Section 7 draws on both to identify priorities for research, policy, and planning. Figure 1 shows the integrated workflow and the points at which the review and spatial analyses

meet. The detailed methodology will be unpacked in the next section.

3. Methods

This review integrates two parallel workflows (figure 1). The integrated research workflow systematically reviews 3625 studies to map what researchers study, where, and how they characterize hazard interactions. From this corpus, we synthesize a hazard interaction taxonomy with decision-tree logic and introduce a triple-mismatch framework to locate convergent deficits in fire burden, research attention, and social vulnerability. The spatial risk workflow analyzes global observational data to map where wildfire-linked risk combinations actually occur across the wildland-to-urban gradient. Together, these workflows enable direct comparison between research attention and physical exposure and identify practical pathways for urban systems across monitoring, prevention, preparedness, response, and long-term resilience.

3.1. Systematic literature review

3.1.1. Search strategy and corpus assembly

A structured search was conducted in the Web of Science (WoS) Core Collection to identify peer-reviewed studies addressing wildfire-linked risks from 2000 to 2025. Search keywords combined wildfire terms with wildfire-linked hazards (e.g. heat, drought, wind, smoke, flooding/debris flow) and multi-hazard interaction terms (e.g. compound, cascading). Full query strings, fields searched, and database indexing settings are provided in supplementary material A.

Retrieved records were exported with title, abstract, keywords, year, and unique identifiers. Duplicate records arising from overlapping query sets were removed, yielding $n = 3625$ unique records for screening.

3.1.2. LLM-assisted classification

The pipeline used OpenAI gpt-4.1 through the application programming interface (API) with a fixed system prompt specifying concise, rule-based classification criteria. Each of the 3625 records was submitted as a separate API call, with the title, abstract, and keywords for that single record provided as input. This per-record processing avoids the context-window limitations and cross-record interference that arise when LLMs are asked to extract structured information from many records simultaneously. The model returned six major outputs:

1. Relevance: whether the study addresses wildfire-linked hazards (excluding unrelated topics such as volcanic eruptions and industrial fires)

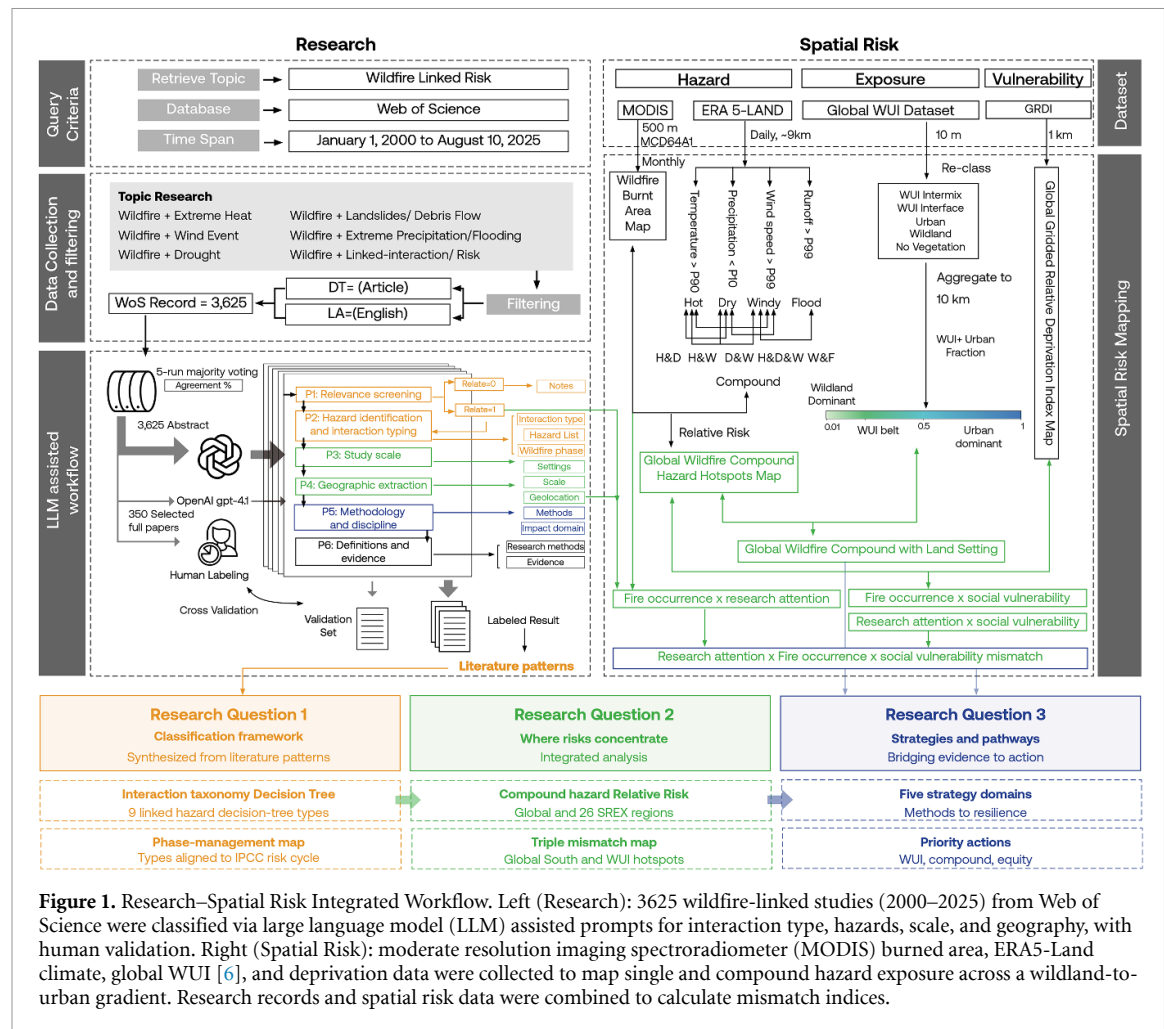


Figure 1. Research–Spatial Risk Integrated Workflow. Left (Research): 3625 wildfire-linked studies (2000–2025) from Web of Science were classified via large language model (LLM) assisted prompts for interaction type, hazards, scale, and geography, with human validation. Right (Spatial Risk): moderate resolution imaging spectroradiometer (MODIS) burned area, ERA5-Land climate, global WUI [6], and deprivation data were collected to map single and compound hazard exposure across a wildland-to-urban gradient. Research records and spatial risk data were combined to calculate mismatch indices.

- Hazard identification and interaction type: which hazards are mentioned and how they interact (using a predefined decision logic described in figure 3)
- Study scale and setting: the spatial extent of analysis (from site-level to global scale) and the land-use setting
- Geographic location: place names, MOB (minimum oriented bounding boxes), and approximate geocenter coordinates of the study area for spatial mapping
- Methodology and discipline: the primary analytical method used and the disciplinary scope of the study
- Definitions and evidence: how authors define and document wildfire-linked relationships

To improve robustness, each record was processed with five independent runs per prompt set, and final labels were assigned by majority vote. Records where the five runs failed to reach majority agreement (below 60% across runs) were flagged for manual review. This ensemble approach reduces sensitivity to LLM variability, following established principles for

improving classification stability through aggregation [23]. Detailed prompt specifications, decision rules, and codebook definitions are provided in supplementary material B2.

3.1.3. Validation

We constructed a targeted evaluation set of 350 records selected from records showing high run-to-run variance. Records whose abstracts contained ambiguous phrasing, mixed settings, or multiple plausible interpretations were selected. These records were manually coded using identical definitions, then compared against aggregated LLM outputs. Overall accuracy was high across all fields (table 1), with mismatch rates ranging from 0.3% for relevance screening to 4.9% for place name extraction. Because this evaluation was applied specifically to the most ambiguous records, mismatch rates in the full corpus where most records are more stable should be lower.

3.2. Global hazard exposure mapping

To quantify where wildfire-linked hazard combinations actually occur, and how patterns differ across urban-wildland settings, we developed a spatial analysis pipeline that integrates multiple earth observation and reanalysis datasets.

Table 1. Agreement between LLM-extracted fields and human validation ($n = 350$) of the highest run-to-run variability literature.

Field	Description	Mismatch (%)
Place names	Geographic location of study area	4.9
Hazard sequence	Ordered list of hazards in causal chain	4.3
Hazard list	Unordered list of hazards mentioned	3.7
Interaction type	Classification per decision tree	3.4
Fire phase	Temporal stage (pre-fire, during, post-fire)	2.9
Scale	Spatial extent of study	2.3
Setting	Land use context (urban, WUI, wildland)	1.7
Linked label	Single vs multi-hazard framing	1.4
Relevance	Inclusion/exclusion decision	0.3

3.2.1. Data sources

We integrate four global datasets covering fire occurrence, climate extremes, land-use setting, and social vulnerability. These datasets were harmonized to a common projection and resampled to the 10 km analysis grid (preprocessing details in supplementary material C1). The analysis window was 2001–2024, aligned with the availability of the MODIS burned-area record. The four datasets span four orders of magnitude in native resolution. We address this through variable-specific aggregation rules described in section 3.2.4 and through a temporally aggregated relative-risk framework introduced in section 3.2.3.

- Wildfire occurrence: MODIS burned-area product (MCD64A1 Collection 6.1), providing monthly burned-pixel flags at 500 m resolution [24]
- Climate extremes: ERA5-Land daily aggregates for temperature, precipitation, wind, and runoff at approximately 9 km spatial resolution [25]
- Urban-wildland setting: Global 10 m WUI classification distinguishing intermix and interface patterns [6]
- Social vulnerability: Global Gridded Relative Deprivation Index (GRDI) at approximately 1 km resolution, with nearest-neighbor interpolation for missing cells [26]

3.2.2. Climate hazard and compound event identification

To examine whether wildfire occurrence is associated with co-occurring meteorological extremes, we classified each day at each pixel as a single- or compound-hazard day. Classification was based on exceedance of locally derived percentile thresholds. Thresholds were defined as per-pixel percentiles computed from the 2001 to 2024 ERA5-Land record:

- Hot (H): daily maximum 2 m air temperature exceeds the local 90th percentile
- Dry (D): daily precipitation sum falls below the local 10th percentile

- Windy (W): daily wind speed exceeds the local 99th percentile
- Flood (F): daily total runoff exceeds the local 99th percentile

Compound hazard days were identified when multiple conditions co-occurred: Hot-and-Dry (H&D), Hot-and-Windy (H&W), Dry-and-Windy (D&W), Hot-Dry-Windy (H&D&W), and Windy-and-Flood (W&F). These combinations capture both fire-conducive conditions and post-fire cascading hazards. Annual frequency layers (days per year) were computed for each single and compound hazard type. Detailed threshold definitions and rationale are provided in supplementary material C2.

3.2.3. Wildfire-compound hazard association

The association between compound hazards and wildfire occurrence was quantified using relative risk (RR) within a temporally aggregated framework. The procedure proceeds in three stages: definition of the compound exposure mask, construction of annual contingency tables, and meta-analytic pooling of yearly estimates. By separating exposure classification (multi-year cumulative) from fire outcome (annual), this design produces a landscape-scale ecological association between cumulative fire occurrence and cumulative compound exposure, rather than event-level attribution of individual fires to daily climate extremes. Both datasets enter this computation at a common analysis scale (section 3.2.4). RR has been widely used in wildfire-climate research to quantify associations between climate extremes and fire occurrence at coarse spatial scales [27, 28].

The compound exposure mask was constructed once from the multi-year frequency record. For each compound type, annual frequency layers (compound days per year) were summed across the full study period to obtain a cumulative exposure count at each pixel:

$$C_i = \sum_{t=2001}^{2024} c_{i,t}, \quad (1)$$

where $c_{i,t}$ denotes the number of compound days at pixel i in year t . Pixels for which C_i exceeded the 75th percentile of the non-zero distribution within the analysis domain (global land or a specific IPCC Special Report on Extremes (SREX) reference region) were classified as exposed; the remainder were classified as unexposed. Holding the mask fixed across years additionally avoids temporal confounding between exposure classification and fire outcome.

For each year t , fire outcome was treated as binary at the pixel level (one if any MODIS burned-area detection occurred in that year, zero otherwise). A 2×2 contingency table was constructed by crossing fire outcome with the fixed exposure mask, and the year-specific RR was computed as:

$$RR_t = \frac{a_t / (a_t + b_t)}{c_t / (c_t + d_t)}, \quad (2)$$

where a_t , b_t , c_t , and d_t denote the number of pixel-years in year t that are exposed-and-burned, exposed-and-unburned, unexposed-and-burned, and unexposed-and-unburned, respectively. Values of $RR_t > 1$ indicate elevated fire likelihood under compound exposure relative to unexposed pixels in the same year.

Year-specific log-RR estimates were pooled across the 24 year record using DerSimonian–Laird random-effects meta-analysis [29], with associations considered statistically significant when the pooled 95% confidence interval excluded $RR = 1$. RR was computed at three stratification levels: globally across all land pixels, per IPCC SREX region, and per urban-wildland setting category. Full pooling specifications, between-year heterogeneity estimation, and the fall-back procedure for years with insufficient within-year variance are provided in supplementary material C3.

3.2.4. Urban-wildland setting classification

The four input datasets (section 3.2.1) span native resolutions from 10 m (WUI) to 9 km (ERA5). All layers were aggregated to a common 10 km grid using variable-specific rules detailed in supplementary material C1: fire pixels by zonal sum, ERA5 by nearest-cell lookup, GRDI by area-weighted mean with nearest-neighbor gap-filling, and WUI by fractional coverage. The 10 km resolution was chosen to remain at or above the coarsest input resolution (~ 9 km ERA5), allowing fire and climate datasets to enter the relative-risk computation (section 3.2.3) at a common analysis scale.

To classify the wildland-to-urban gradient, we derived a continuous setting index from the urban and WUI fractional coverage in each 10 km cell:

$$S = U + (1 - U) \cdot W, \quad (3)$$

where U is the urban fraction of the cell and W is the WUI fraction among non-urban pixels within the cell. The index ranges from 0 (no built cover) to 1

(entirely built-up), with intermediate values reflecting transitional zones where buildings and vegetation meet or intermingle. We binned S into three categories:

- Wildland-dominant: $S < 0.01$, predominantly undeveloped wildland
- WUI belt: $0.01 \leq S < 0.5$, transitional buildings-and-vegetation zone
- Urban-dominant: $S \geq 0.5$, dominantly built-up with secondary WUI presence

This classification allows compound hazard RR to be stratified by setting category. The setting index formulation and threshold rationale are detailed in supplementary material C4.

3.2.5. Research-exposure-vulnerability mismatch indices

To quantify gaps between research attention and actual hazard risk exposure, we constructed four composite indices on a regular $2^\circ \times 2^\circ$ global grid. Each cell was characterized by three components: fire burden (cumulative MODIS burned-area pixel count from MCD64A1, 2001–2024), social vulnerability (mean GRDI, with nearest-neighbor interpolation for missing cells), and research attention.

Research attention was allocated to grid cells in proportion to the geographic footprint of each study, following the bounding-box geoparsing approach of Lin *et al* [30], who applied this method to more than 60 000 wildfire studies at 1° resolution. For a study with K geoparsed regions, each region contributes weight $1/K$ distributed across overlapping land cells in proportion to their cos-latitude corrected area of intersection. Every study therefore contributes total weight 1.0 across the global land surface.

Each component was rescaled to its percentile rank, following standard composite index practice [31–33]. The three pairwise indices (fire $\times$ research, fire $\times$ vulnerability, vulnerability $\times$ research) and the triple mismatch index were constructed as geometric means of the ranked components, so that high scores arise only when all constituent dimensions are simultaneously elevated. The triple index additionally applies a convergence penalty to cells where any single dimension falls below the 25th percentile, isolating areas where all three deficits coincide rather than areas where any single deficit is severe. Full formulas and normalization steps are in supplementary material C5.

Sensitivity to the choice of analysis-grid resolution was tested at 0.5° , 1° , 2° , and 5° . Spearman rank correlations of SREX-region mean mismatch scores exceed 0.97 between any pair of resolutions for all four indices, and the top-10 most-mismatched regions are stable across resolutions (supplementary material C5.9). We use the 2° baseline in this paper.

4. Wildfire-linked risk in the literature: research evolution and definitions

4.1. Timeline and trends in wildfire-linked risk literature

Publication counts on wildfire-linked hazards increased sharply from 2000 to 2024, with acceleration after 2015 (figure 2). Interaction-type labels are defined with examples and management implications in table 2, and the decision tree used to assign them is shown in figure 3. The timing aligns with successive IPCC assessment cycles. Modest growth followed the Fourth (AR4) and Fifth (AR5) IPCC Assessment Reports [34, 35], while steeper acceleration coincided with the Special Reports on 1.5 °C warming, climate and land, and oceans and cryosphere, as well as AR6 [8, 36]. This pattern reflects broader institutional emphasis on compound risk and extreme-event attribution [37, 38].

Despite this growth, studies on different hazard types remain unevenly distributed across the corpus. Single-hazard studies are most common (32.1%), indicating that much wildfire research still examines fire as an isolated phenomenon. Among studies focusing on multi-hazards, preconditioned and triggered relationships each account for 13.5% of studies, together representing more than a quarter of the corpus. Concurrent and compound interactions constitute 8.9% of the full corpus. The remaining types (cascading, antagonistic, temporally compounding, spatially compounding, and amplified) collectively account for less than 10% of the corpus. This distribution suggests that while multi-hazard dynamics receive increasing attention, the literature still remains largely single directional in framing: studies focus either on conditions that set the stage for fire, or on hazards that follow it. Concurrent, compounding, and cascading dynamics are far less studied.

Thematic clustering between hazard families and interaction types reveals disciplinary patterns (figure 2, chord diagram). Preconditioning links primarily to heat and drought; triggered and cascading framings associate with post-fire geomorphic hazards; air-quality hazards connect notably to concurrent framing. These clusters indicate that different research communities favor particular terminology. Fire ecologists speak of drought-fire preconditioning, geomorphologists of fire-debris flow triggering, and environmental health researchers of smoke-heat concurrence. The uneven distribution of interaction labels thus reflects not only research trends but also the absence of a shared classification framework, a limitation this review addresses in the following section.

4.2. Hazard interaction terminology

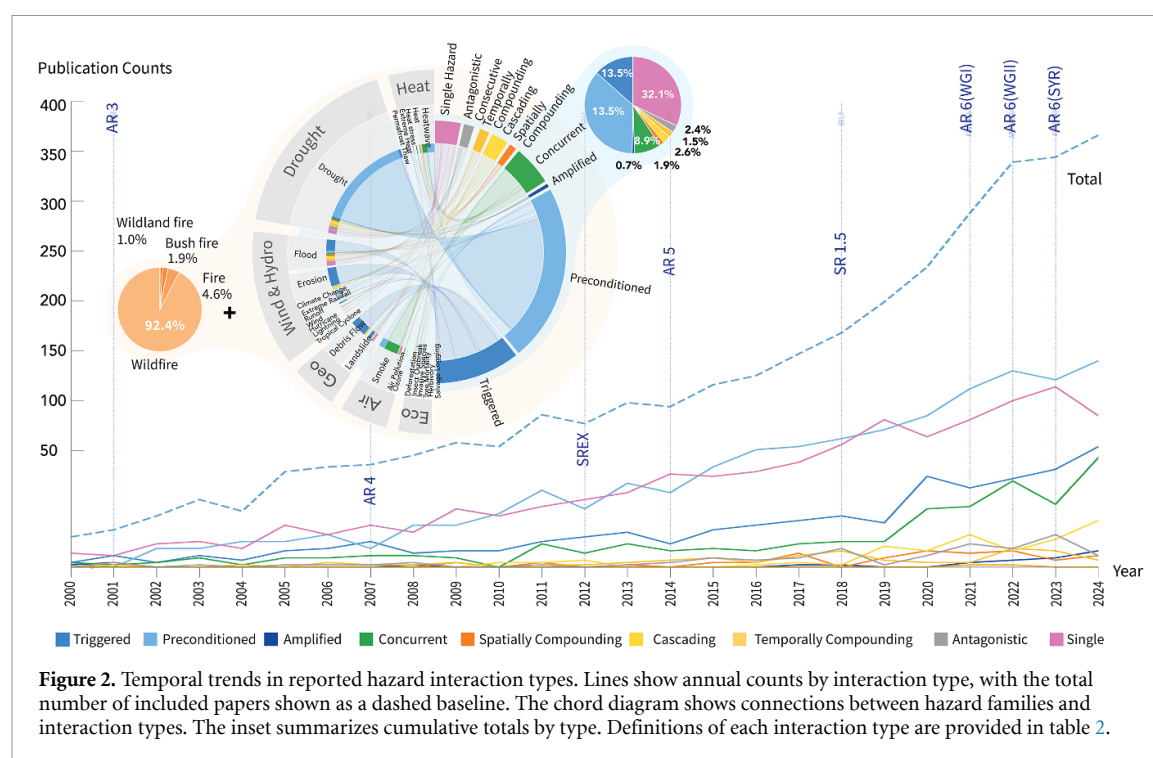
The language used to describe multi-hazard wildfire risk has evolved substantially over the past decade,

driven by successive IPCC assessment cycles and parallel conceptual work. The SREX report (2012) introduced multi-hazard perspectives for adaptation and emphasized that risk emerges from interacting drivers rather than isolated events [9]. AR5 expanded attention to compounding processes, and AR6 consolidated the compound event concept, defining it as ‘the combination of multiple drivers and/or hazards that contributes to societal and/or environmental risk’ [8]. AR6 also emphasized cascading impacts, recognizing that risks propagate through connected systems rather than remaining confined to a single hazard event [36].

In parallel, Zscheischler *et al* synthesized a widely adopted typology organizing compound events into four categories: preconditioned, multivariate (concurrent), temporally compounding, and spatially compounding [12]. This framework has become a standard reference across climate and hazard research [11, 18]. However, applying it to wildfire in urban settings reveals a gap. The four categories capture co-occurrence and accumulation well, but they do not explicitly distinguish directed causal sequences (where one hazard triggers another) from broader systemic cascades (where impacts propagate through infrastructure networks). This distinction is critical for urban hazard management, where the relevant question is often not ‘did these hazards overlap?’ but ‘did one cause the next, and through what system?’

To address this gap, we developed a decision tree (figure 3) that extends the Zscheischler *et al* typology with an explicit causal-chain branch. The tree uses a small set of hierarchically organized questions applied in fixed order: sign (amplifying or alleviating?), spatial extent (same or different location?), temporal structure (does one event follow another, and if so, is the chain multi-step, accumulative, or back-to-back?), co-occurrence (simultaneous but non-causal?), and finally causal direction (does one hazard directly produce, precondition, or amplify the next?). This structure reduces ambiguity and makes coding decisions auditable across the full corpus.

The causal-chain branch distinguishes two interaction types that existing typologies conflate. *Triggered* interactions describe cases where fire directly increases the likelihood of a specific subsequent hazard, such as post-fire rainfall producing debris flows in burned watersheds [3, 39]. *Cascading* interactions describe multi-system propagation, in which impacts pass through infrastructure nodes and institutional boundaries before reaching their ultimate consequences [19, 20]. AR6 Working Group II (WGII) recognizes both processes in urban contexts, noting that cities face compounding and cascading risks from hazard interactions and infrastructure interdependencies [17]. Our decision tree operationalizes this recognition for systematic classification.



4.3. Interaction types map to distinct management phases

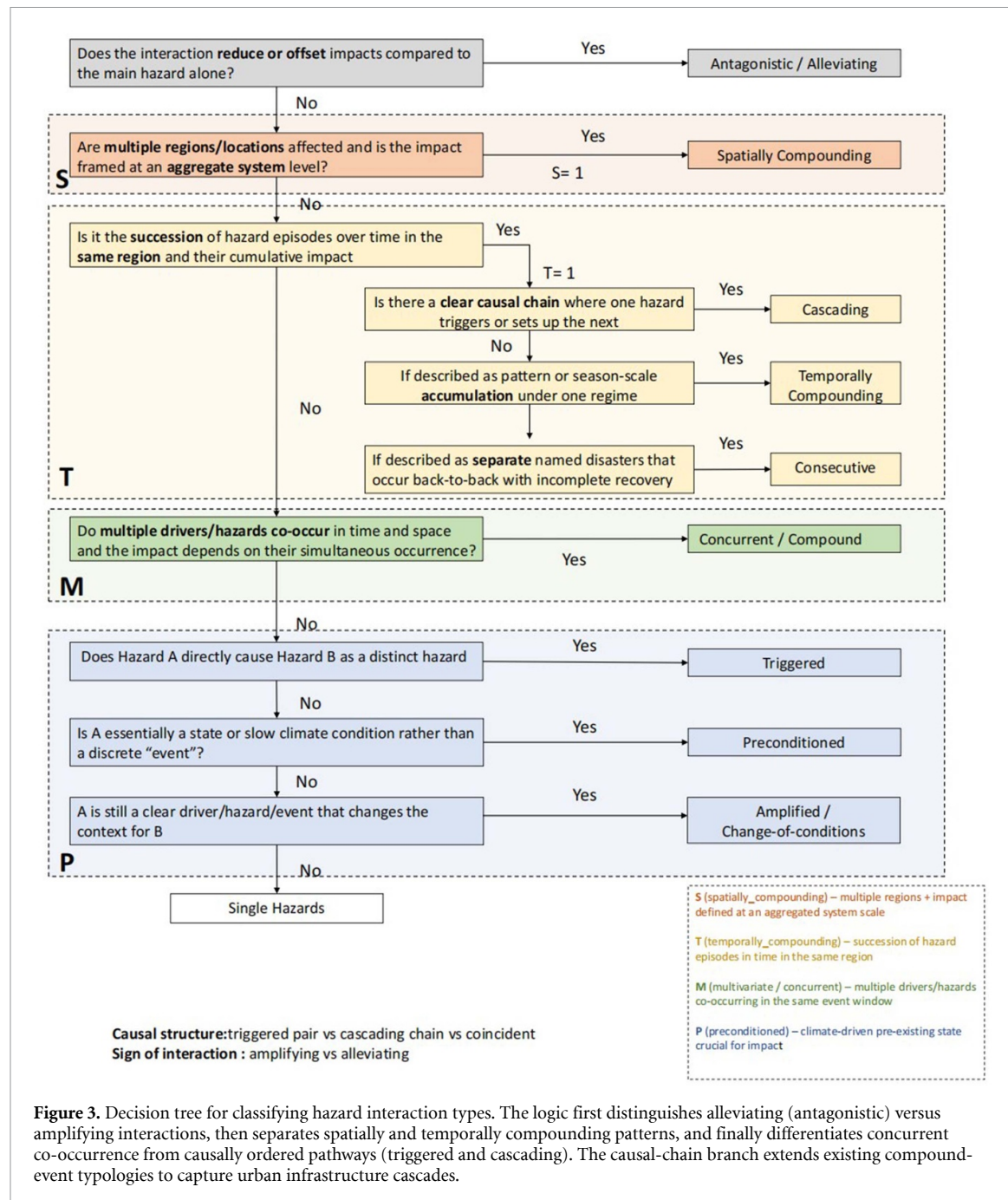
Each interaction type imposes distinct demands on response strategy, timescale, and lead agency. A drought that preconditions fire severity over months presents a markedly different management challenge than a debris flow triggered hours after rainfall on burned slopes. In IPCC risk frameworks, wildfire-linked risk emerges from the intersection of hazard, exposure, and vulnerability [8]. Each interaction type modifies these components differently and aligns with distinct phases of the disaster management cycle (figure 4). Table 2 summarizes the full set of interaction types, definitions, and management implications.

Prevention and mitigation: Some interactions unfold over months to years, offering long lead times for intervention. Preconditioning is the clearest example: multi-season drought reduces fuel moisture and primes landscapes for severe fire [14, 46], enabling anticipatory measures such as drought monitoring, fuel treatment scheduling, and development restrictions [5, 47]. Amplified interactions work through a shifting baseline. The documented increase in compound fire-weather episodes renders historical thresholds unreliable [28, 41], meaning prevention strategies calibrated to past conditions may systematically underestimate future risk. Triggered interactions also inform mitigation at this timescale: watershed treatments that reduce post-fire debris flow susceptibility must begin before the triggering fire occurs [3, 48].

Preparedness: For preconditioned interactions with long lead times, seasonal climate forecasts enable

anticipatory resource allocation [46]. For triggered interactions, the critical period is the lag window between fire and its downstream consequences. The 2018 Montecito debris flow illustrates this: weeks after the Thomas Fire burned steep watersheds, a single rainstorm mobilized debris that killed 23 people and destroyed over 100 homes [3, 4]. The interaction type (triggered, not concurrent) determined who needed to act and when: geotechnical agencies in the days to weeks after containment, not fire suppression teams during active burning. Temporally compounding interactions require different preparedness logic. Australia's 2019–20 'Black Summer' produced prolonged, severe smoke episodes across eastern Australia [2], requiring cumulative exposure tracking and staged response protocols rather than single-event triggers.

Response: Concurrent and spatially compounding interactions pose the most acute response challenges because they compress decision timelines and overwhelm single-hazard systems. When wildfire smoke degrades air quality during extreme heat, evacuation becomes hazardous, shelters must provide both cooling and air filtration, and public health messaging must address compound exposures [42, 49]. Emergency managers trained to activate heat plans or smoke advisories separately face compound scenarios that neither plan anticipates. Hospitals receive simultaneous surges in respiratory and heat-related admissions while staff availability declines [50, 51]. Spatially compounding interactions create a different crisis: when fire danger synchronizes across regions, mutual aid networks fail because neighboring jurisdictions face simultaneous emergencies [43, 44]. The



2021 Marshall Fire demonstrated how wind-driven fire in a suburban setting can overwhelm local capacity within hours [52]. Under these conditions, the management response shifts from resource deployment to resource triage.

Recovery: Cascading interactions propagate through interconnected urban systems, transforming localized infrastructure damage into regional crises. When wildfire damages power infrastructure, the failure propagates: water pumping ceases, telecommunications degrade, and medical equipment loses power [17, 19, 20, 53]. No single agency has jurisdiction over this entire chain, and recovery requires interdependency mapping and cross-sector coordination that current arrangements rarely

provide. Consecutive interactions compound the challenge: communities experiencing repeated fire seasons face cumulative erosion of housing stock, economic base, and institutional capacity before recovery from the previous event is complete [45]. Building resilience under these conditions requires land use changes that reduce re-exposure, insurance mechanisms that maintain coverage, and governance structures that sustain recovery investment across budget cycles [12].

Finally, antagonistic interactions deserve attention precisely because they are often overlooked. Wetting rainfall that suppresses fire activity represents a protective window [40], and identifying when and where alleviating conditions occur can inform

Table 2. Interaction type definitions and management implications.

Interaction type	Definition	Example	Management phase
Antagonistic / Alleviating Preconditioned	Conditions dampen wildfire impacts or probability Slow background states set the stage for severe outcomes	Wetting rain days suppressing fire activity [40] Multi-season drought elevating fuel aridity [14]	Prevention: identify protective windows Prevention, Mitigation: anticipatory monitoring
Amplified / Change-of-conditions Concurrent / Compound	A distinct driver shifts operating context, increasing fire probability or severity Simultaneity of multiple hazards; risk depends on joint exposure	Increasing compound fire-weather episodes [28, 41] Extreme heat combined with wildfire smoke [42]	Prevention: scenario planning; updated thresholds Response: integrated operations and messaging
Spatially Compounding	Fire danger becomes geographically synchronous across regions	Multiple large fires straining suppression capacity [43, 44]	Response: mutual aid networks; resource triage
Temporally Compounding	Repeated episodes or season-long accumulation	Persistent smoke exposure over fire season [7]	Preparedness: cumulative exposure tracking
Consecutive Triggered	Back-to-back shocks with incomplete recovery Fire directly increases likelihood of subsequent hazard	Sequential fire seasons affecting same communities [45] Post-fire debris flows following rainfall [3]	Recovery: resilience building Preparedness: lag-window monitoring
Cascading	Multi-system propagation through infrastructure and services	Power outage affecting water pumping and emergency response [17, 19, 20]	Recovery: interdependency management

the timing of prescribed burns, infrastructure maintenance, and preparedness activities.

5. Global fire patterns, compound drivers, and the triple mismatch

5.1. Compound climate conditions shape wildfire risk along a fire regime gradient

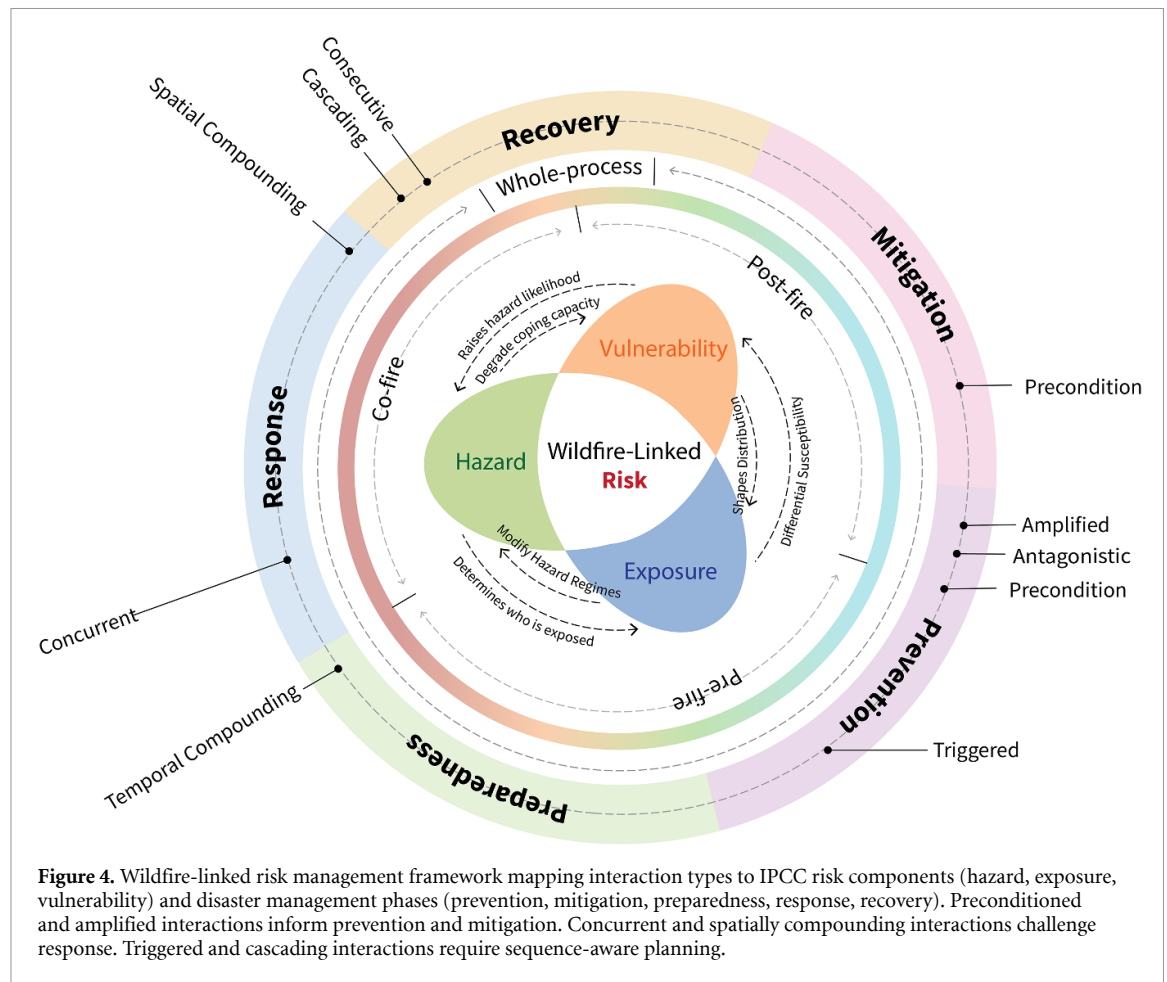
Compound climate conditions are increasingly recognized as strongly associated with wildfire occurrence. Their frequency has increased globally over the past four decades [28], and climate change has raised the probability of extreme fire years by 88%–152% across global forested lands [27]. Yet most assessments rely on aggregate fire weather indices that combine multiple variables into a single score [54]. While these indices provide operational convenience, they cannot identify which specific variable combinations elevate risk or whether those relationships differ across fire regimes.

This distinction has become increasingly relevant as climate adaptation policy turns toward regionally specific risk reduction. Region-specific adaptation requires knowing not only that fire risk is rising, but which mechanism is driving the rise in each region. We therefore decompose compound conditions and analyze them at both global and regional scales to address two questions. The first concerns the gap between global and regional risk patterns. A globally averaged measure of compound fire weather can mask substantial regional variation in which specific

compound combinations matter most, as the contrast between hotspot maps using global and regional thresholds shows (figures 5(a) and 6(a)). The same global trend may reflect very different mechanisms in different parts of the world. The second concerns mechanism identification at the regional scale. Knowing that fire risk is rising in a given region is less actionable than knowing which compound combination is driving the rise, because adaptation strategies differ depending on whether the dominant pathway is drought, wind-coupling, or moisture-pulse driven.

Our global analysis confirms the dominance of compound conditions (figure 5(b)). Threshold definitions for each single and compound class are given in section 3.2.2. Hot-and-dry conditions show the strongest association with wildfire occurrence ($RR = 4.87$), followed by dry-and-windy ($RR = 4.56$), hot-dry-windy ($RR = 3.15$), and hot-and-windy ($RR = 2.68$). Single extremes are weak predictors at the global scale. These results are consistent with prior findings that burned area increases most when fire weather compounds with dry antecedent conditions [14, 41].

These global averages mask sharp regional contrasts that are rooted in distinct fire regime types (figure 6; full SREX region values in supplementary material D1). To make these regional patterns visible, we recomputed compound hotspots using region-specific (per-SREX) percentile thresholds (figure 6(a)). This re-calibration brings out locally extreme events that the globally uniform thresholds in figure 5(a) do not capture. We follow the global classification of [55], who used Bayesian



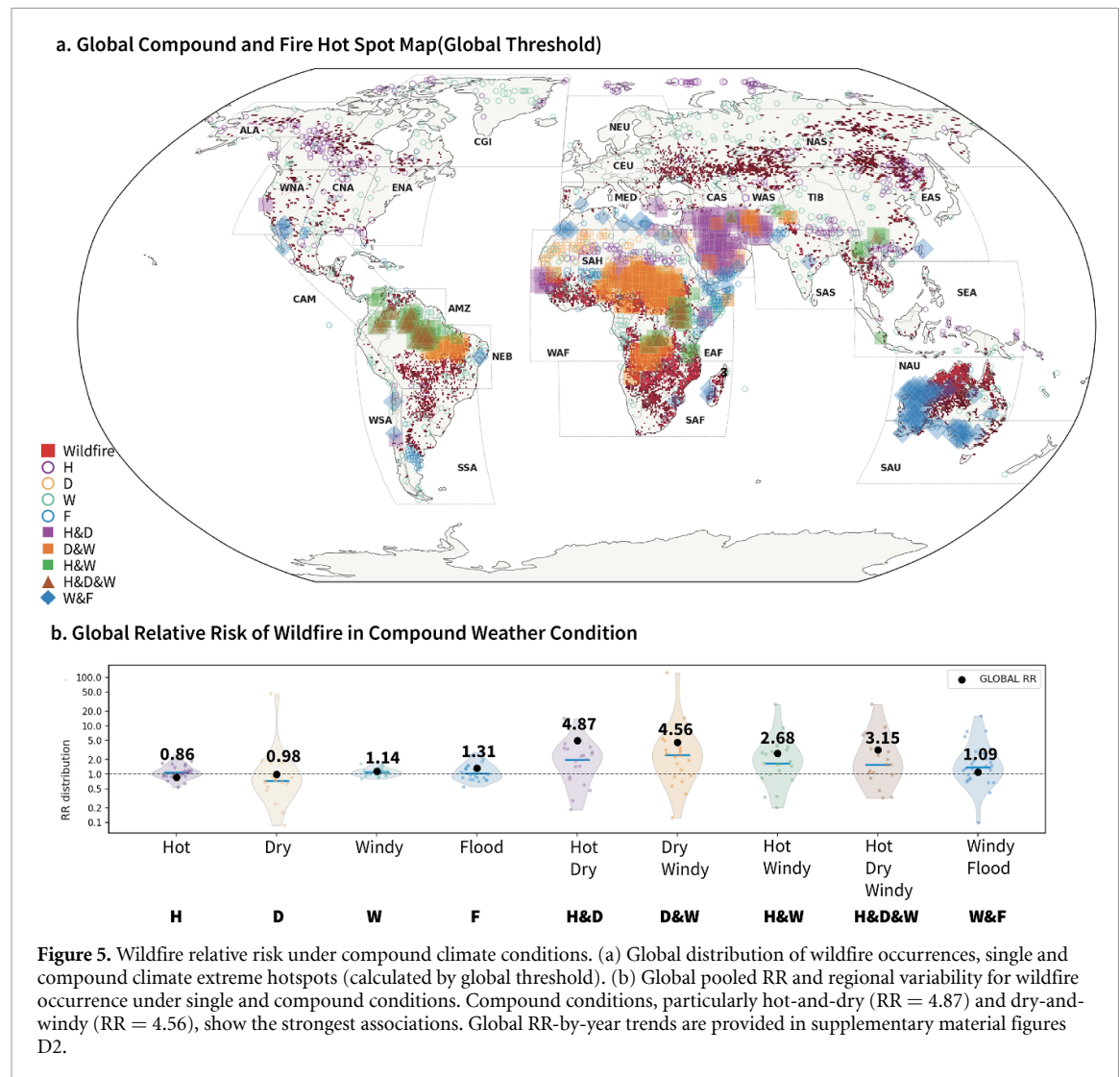
inference on GFED4s burned area observations to partition land into regions dominated by fuel, moisture, and ignition controls. We grouped the 26 SREX regions into four fire regime categories: fuel-limited, moisture-limited, ignition-limited, and intermediate productivity. Category assignments were made by visual inspection of the dominant control class in Kelley *et al*'s global potential limitation map (Supplementary figure D1). Figure 6(b) shows two focal regions per category, selected as the clearest examples.

In fuel-limited systems, the atmosphere is nearly always conducive to burning, but fire is constrained by the availability and continuity of vegetation. In moisture-limited systems, fuel loads are sufficient, yet vegetation and atmospheric moisture remain too high for combustion except under extreme weather [56, 57]. In ignition-limited systems, fire is constrained by the scarcity of natural and human ignition sources despite favorable fuel and climate. Between these primary regime types, intermediate-productivity landscapes (savannas and many Mediterranean-type ecosystems) experience the highest fire activity globally, consistent with the intermediate productivity hypothesis [58, 59]. The compound pathways identified in our analysis

(figure 6) follow this pattern: drought-driven compounds dominate where moisture limits burning, while wind-and-flood compounds become critical where fuel is limiting, and multi-factor alignment is required where ignitions are limiting.

The clearest illustration of fuel limitation appears in the arid interiors of Central and West Asia. In Central Asia, all heat- and drought-containing compounds are associated with significantly reduced fire likelihood (e.g. hot-and-dry $RR = 0.29$; dry-and-windy $RR = 0.46$). Only windy-and-flood conditions elevate risk ($RR = 2.19$), suggesting that episodic moisture pulses temporarily restore fuel continuity. When wind is also present, fire can then spread. West Asia shows an even stronger version of this pattern, with dry-and-windy reaching $RR = 0.17$. These results, where RR falls substantially below 1.0 under conditions that elevate risk elsewhere, follow directly from fuel-limitation logic: where extreme aridity prevents vegetation accumulation, additional drying removes the fuel needed to carry fire [58, 60].

In moisture-limited fire regimes, fire is rare under normal conditions because high fuel moisture suppresses flammability. Drought-containing compounds become the primary pathway by which fire penetrates these systems. Southeast Asia provides a

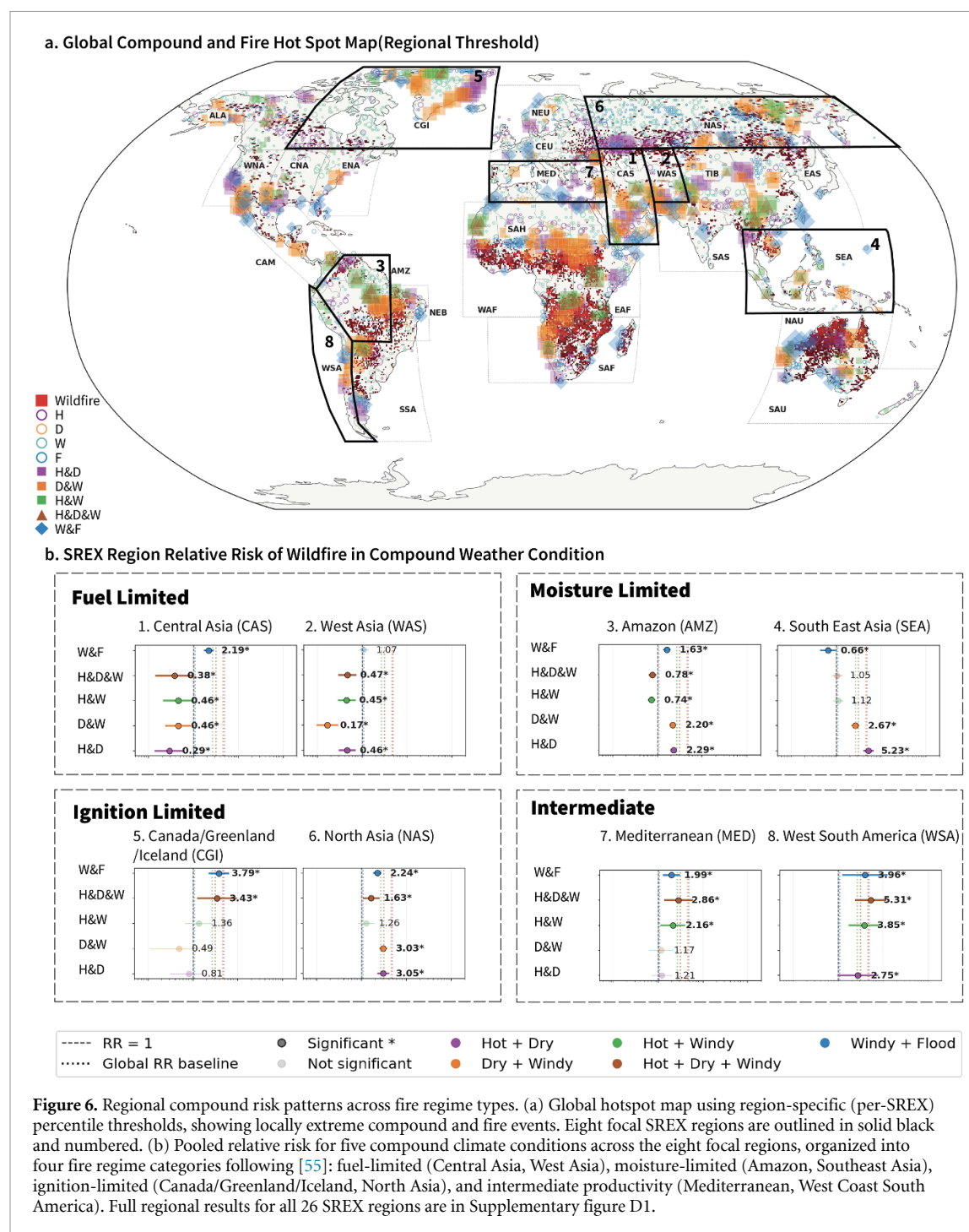


clear example: hot-and-dry ($RR = 5.23$) and dry-and-windy ($RR = 2.67$) conditions strongly elevate risk, while windy-and-flood reduces risk ($RR = 0.66$). The Amazon shows a similar pattern (hot-and-dry $RR = 2.29$; dry-and-windy $RR = 2.20$), consistent with evidence that below-average precipitation enables fire spread to forests where it is historically rare [61]. Yet adding wind to hot-and-dry conditions in the Amazon reverses the direction of the association (hot-dry-windy $RR = 0.78$, indicating a protective effect). In this region, strong winds are often associated with convective storm systems that bring rainfall, suppressing flammability [62].

In ignition-limited systems, neither dry nor ignition-enabling conditions alone are sufficient to overcome the combined constraint, but compound alignment across multiple factors can enable fire spread. Northern Asia shows four of five compounds significantly elevated (hot-and-dry $RR = 3.05$; dry-and-windy $RR = 3.03$). The Canada-Greenland-Iceland region shows a distinctive variant in which dry compounds alone

show a non-significant protective tendency (dry-and-windy $RR = 0.49$), but compound alignment strongly elevates fire (windy-and-flood $RR = 3.79$; hot-dry-windy $RR = 3.43$). The projected increase in lightning strikes under climate warming [63] is therefore particularly consequential for these regions, because it acts on the specific constraint that currently suppresses their fire activity.

The most complex compound behavior, and the most extreme effect sizes, appear in landscapes at intermediate productivity, where the dominant limiting factor shifts seasonally. Two patterns stand out. First, ordinarily moist environments produce the largest risk amplification when compound conditions align. West Coast South America shows the largest amplification of any focal region (hot-dry-windy $RR = 5.31$; hot-and-windy $RR = 3.85$). The Mediterranean reaches $RR = 2.86$ for hot-dry-windy and $RR = 2.16$ for hot-and-windy, while hot-and-dry alone is close to 1 ($RR = 1.21$), consistent with wind-driven fire behavior documented during the

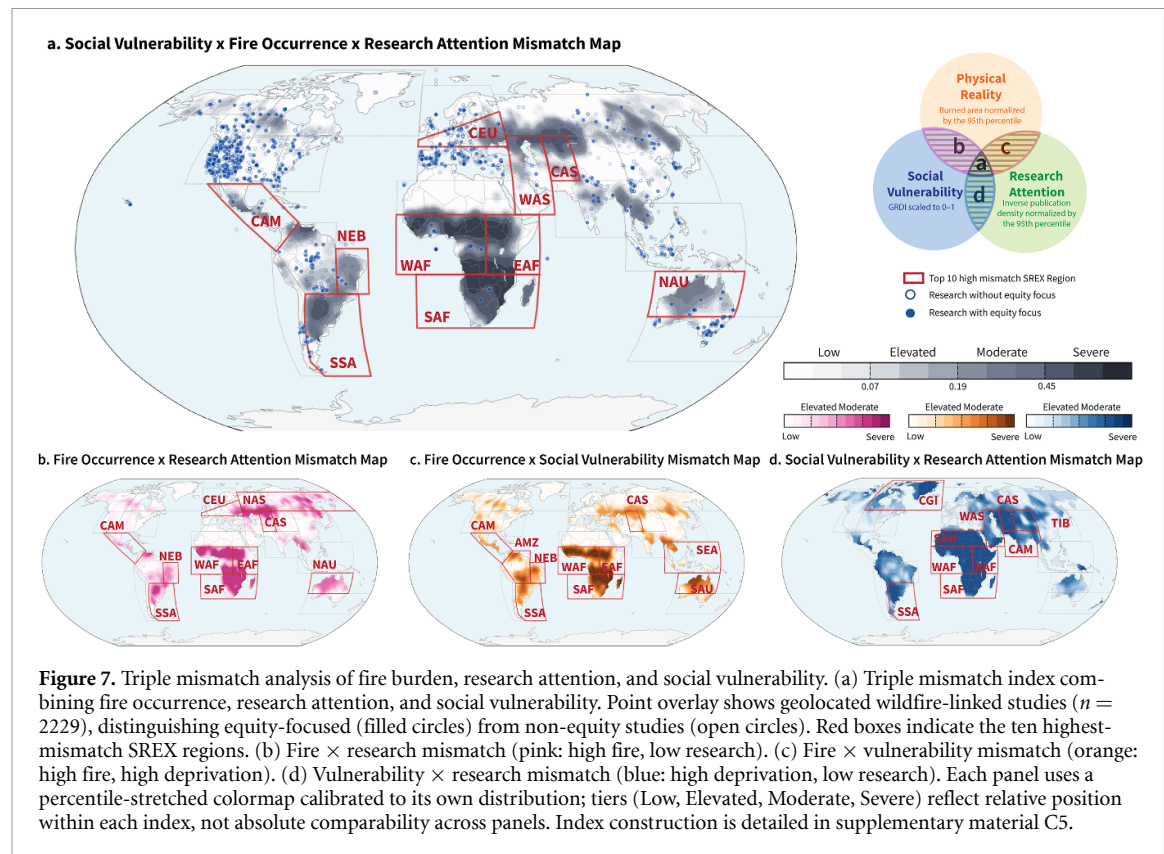


2020 western Oregon fires [43]. These extreme values reflect the rarity of compound fire weather in wet climates. When conditions do converge, the risk amplification is disproportionately large.

Second, the leading compound pathway varies even within the intermediate category, as illustrated by the contrast between Mediterranean and West Coast South America in their dominant compound combination. A single compound index would therefore misidentify the dominant pathway across these regions, consistent with the role of fuel connectivity in seasonally productive landscapes, where wet-season rainfall builds fuel loads and dry-season conditions

determine which pathway dominates ignition and spread [64].

Taken together, these patterns form a coherent picture. Drought-containing compounds dominate in moisture-limited systems, overcoming the moisture constraint on flammability. Drought suppresses fire in fuel-limited systems by reducing vegetation, and wind-and-flood compounds elevate risk instead. Ignition-limited systems require compound alignment across multiple factors. Intermediate-productivity systems show the most variable signatures and the largest amplification under compound alignment. Year-by-year global RR estimates



show that compound conditions consistently elevate fire risk across the 24 year record, with a stable tier structure among compound types (supplementary material figure D2). For risk assessment, these contrasts mean that monitoring systems must be adapted to the local compound structure and underlying fire regime rather than relying on globally uniform indices.

5.2. The triple mismatch: fire burden, research neglect, and social vulnerability

The compound risk patterns described above are shaped by deep inequities in research coverage and social vulnerability. We examine the convergence of three dimensions, fire burden, research neglect, and social vulnerability, into a triple-mismatch framework, then decompose this convergence into three pairwise indices (figure 7). Research attention is allocated to grid cells using cos-latitude weighted area of intersection between each study's geoparsed bounding boxes and the analysis grid (supplementary material C5.2), so every study contributes total weight 1.0 across the global land surface regardless of spatial scope. We define cells as 'high-mismatch' when their index value is at or above the panel-specific 75th percentile of the nonzero distribution; this uniform threshold allows direct comparison across the four indices. These rankings are robust to grid choice: Spearman rank correlations between the 2° baseline and grids at 0.5° , 1° , and 5° exceed 0.97 for every pairwise and composite index, with Jaccard overlap of

0.82 to 1.00 for the top-10 SREX regions across resolutions (supplementary material C5.9).

The triple mismatch index identifies where all three dimensions converge (figure 7(a)). The sub-Saharan savanna belt scores highest globally, with West Africa, Southern Africa, and East Africa as the leading SREX regions. Southeast South America, Central Asia, Northeast Brazil, Northern Australia, and Central America and Mexico form the second tier. Central Europe and North Asia also score elevated, where lower deprivation is partly offset by elevated fire burden and limited research coverage. No region in the Global North ranks in the top eight of the triple mismatch index. The distribution of study locations reinforces this pattern: the vast majority of studies cluster in North America, Europe, and southeastern Australia, while many nations with regular fire exposure in the Global South have fewer studies than localized fire-prone areas of California or Mediterranean Europe, reflecting structural inequities in research funding and institutional capacity [30].

The three pairwise indices decompose this convergence and reveal which dimension dominates in different regions. The fire-research mismatch (figure 7(b)) shows that research coverage is sharply misaligned with fire activity across the Global South, with a pronounced hemispheric contrast: research clusters in western North America, the Mediterranean, and eastern Australia, while large fire-prone regions of Africa, South America, and

Southeast Asia receive minimal coverage [5, 65]. The three highest-ranked SREX regions are West Africa, Southern Africa, and East Africa, followed by Northern Australia, Southeast South America, and Northeast Brazil. The fire-vulnerability mismatch (figure 7(c)) again identifies a belt of convergent fire burden and deprivation across sub-Saharan Africa, with West Africa, Southern Africa, and East Africa leading the rankings. Here, extensive annual burning overlaps with populations facing constrained access to healthcare, warning systems, and post-disaster support. Northeast Brazil, the Amazon basin, and Southeast Asia rank close behind, where peatland and agricultural fires intersect with rural poverty [5, 13]. The vulnerability-research mismatch (figure 7(d)) is the most geographically widespread of the three: 5 SREX regions reach the Severe tier (Central Asia, East Africa, Sahara, Southern Africa, West Africa) and 13 of 26 regions sit at Moderate or higher, indicating that socially vulnerable populations across most regions lack research coverage proportional to their need. The dominant contribution comes from arid and semi-arid Eurasia (Central Asia, Sahara, South Asia, Tibetan Plateau, and West Asia) where deprivation is high but the published wildfire literature is sparse, with East Africa, West Africa, Southern Africa, and Southeast South America extending the pattern across the Global South.

Together, these geographic patterns carry three implications for how the field allocates attention and resources. First, the mismatch represents an allocation failure: the global fire science enterprise invests least where fire burns most and communities are most exposed. Second, populations facing compounded fire and deprivation are largely absent from the evidence informing adaptation guidance, risk standards, and funding priorities [8]. Third, what limited equity research exists concentrates in moderate-deprivation regions where adaptive capacity is relatively high, rather than in the most-deprived areas where fire interacts with extreme vulnerability [21, 22].

6. Wildfire-linked hazards in urban systems

6.1. Compound-dependent fire risk across the wildland-to-urban gradient

Within our analysis, wildland areas contain negligible built cover, WUI areas are the transitional belt where buildings and wildland vegetation meet or intermingle, and urban areas are dominantly built-up (section 3.2.4). The global mismatch documented in figure 7 concentrates most severely at the WUI, where lack of research attention, fire exposure, and social vulnerability converge. Of the 3625 studies in our corpus, 96.0% focus on wildland settings, while WUI belts account for 0.9% and urban areas for 3.1%. Yet wildland areas account for only 66.8% of

actual fire occurrence; WUI areas contribute 28.7% and urban areas 4.5% (figure 8). Fire risk in built-up settings operates through mechanisms that wildland-focused research has not characterized. Even within wildland research, studies are geographically concentrated. Many fire-active wildland regions, particularly across the Global South, remain sparsely studied, which keeps the wildland mismatch score elevated.

Figure 8 shows that the proportion of fire occurrence, social deprivation, and research attention shifts substantially across the wildland-to-urban gradient, yet existing compound-event frameworks treat these settings as equivalent [12, 15]. The gradient restructures every component of risk: fuel loads decline while infrastructure density rises, population exposure shifts from dispersed to concentrated, and adaptive capacity differs qualitatively. Treating all settings as equivalent obscures the distinct vulnerabilities that each presents.

These distributional shifts have a direct mechanistic counterpart: urban and wildland settings respond to climate extremes through distinct mechanisms (figure 9). In wildland contexts, single extremes produce modest risk elevation. Compound conditions raise risk broadly across multiple hazard combinations. This profile is consistent with heat, aridity, and wind acting on continuous fuels [13, 14]. Urban areas display a sharply different sensitivity: single extremes produce negligible risk elevation, yet compound conditions produce the strongest amplification of any setting. Hot & dry and dry & windy combinations reach RR values of approximately 5.4 and 5.3, respectively.

This sharp, disproportionate increase indicates that urban fire risk is predominantly compound-dependent, emerging from the co-occurrence of multiple stressors rather than from any single climate driver. Several mechanisms likely contribute: urban heat island effects that intensify fire weather, infrastructure-mediated ignition under compound stress (such as power line failures during extreme wind events), and reduced moisture buffering in built environments. Their relative importance remains unquantified [52, 62, 66]. The WUI belt sits between these two settings. It shows strong compound amplification but intermediate effect sizes, reflecting its transitional fuel structure [6].

The compound-dependent risk profile coincides with a systematic misalignment between where fires occur, where vulnerable populations live, and where research concentrates (figure 8). We quantify this misalignment across three pairwise dimensions and one combined index. Fire occurrence and research attention are misaligned in all settings, but the WUI scores highest (0.684 vs 0.681 wildland and 0.654 urban; figure 8(e)). Fire occurrence and social vulnerability also diverge most in the WUI (0.555 vs 0.532 wildland and 0.456 urban; figure 8(f)), consistent with affordable housing and marginalized communities clustering in fire-prone zones [5, 21, 67].

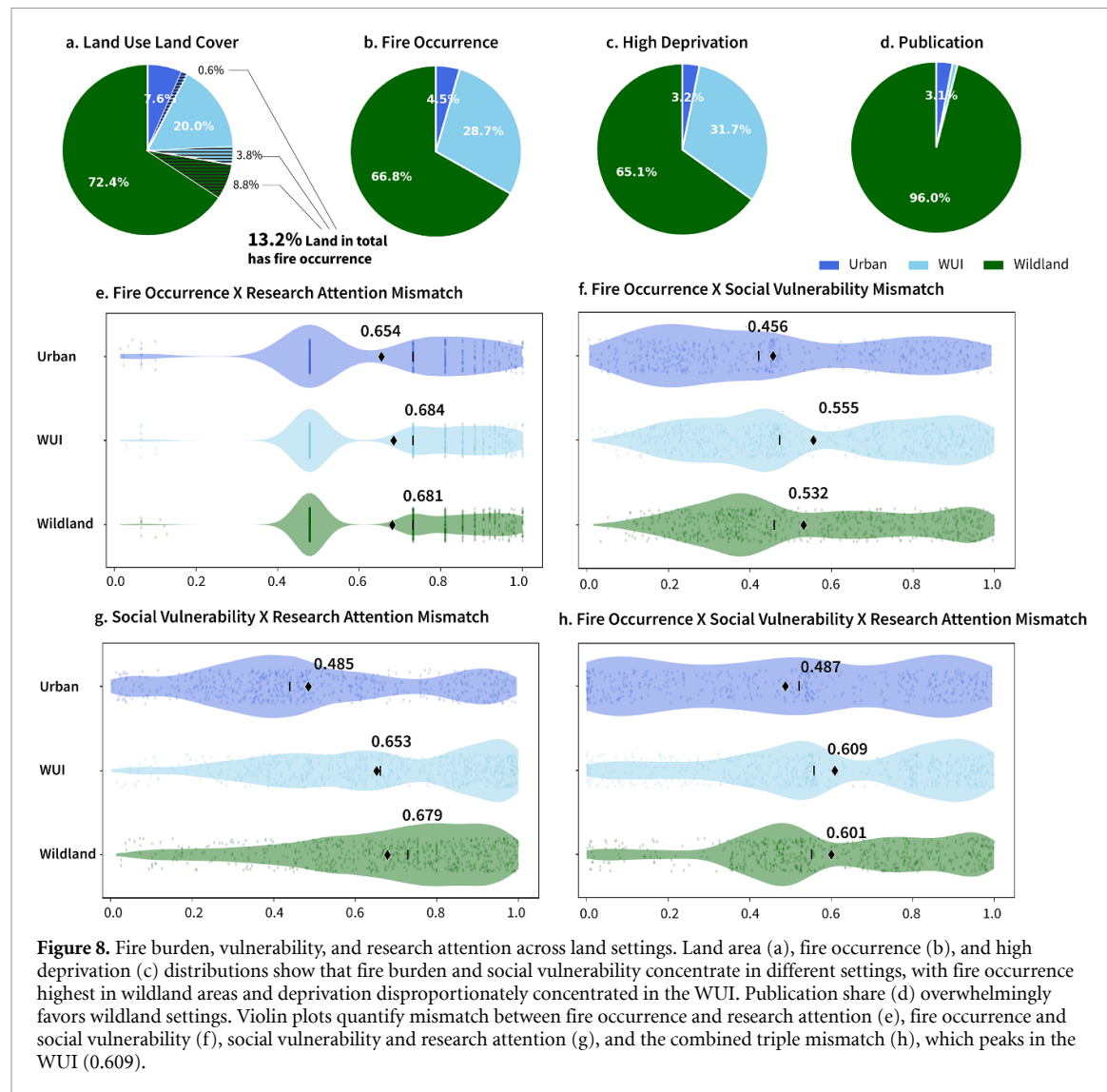


Figure 8. Fire burden, vulnerability, and research attention across land settings. Land area (a), fire occurrence (b), and high deprivation (c) distributions show that fire burden and social vulnerability concentrate in different settings, with fire occurrence highest in wildland areas and deprivation disproportionately concentrated in the WUI. Publication share (d) overwhelmingly favors wildland settings. Violin plots quantify mismatch between fire occurrence and research attention (e), fire occurrence and social vulnerability (f), social vulnerability and research attention (g), and the combined triple mismatch (h), which peaks in the WUI (0.609).

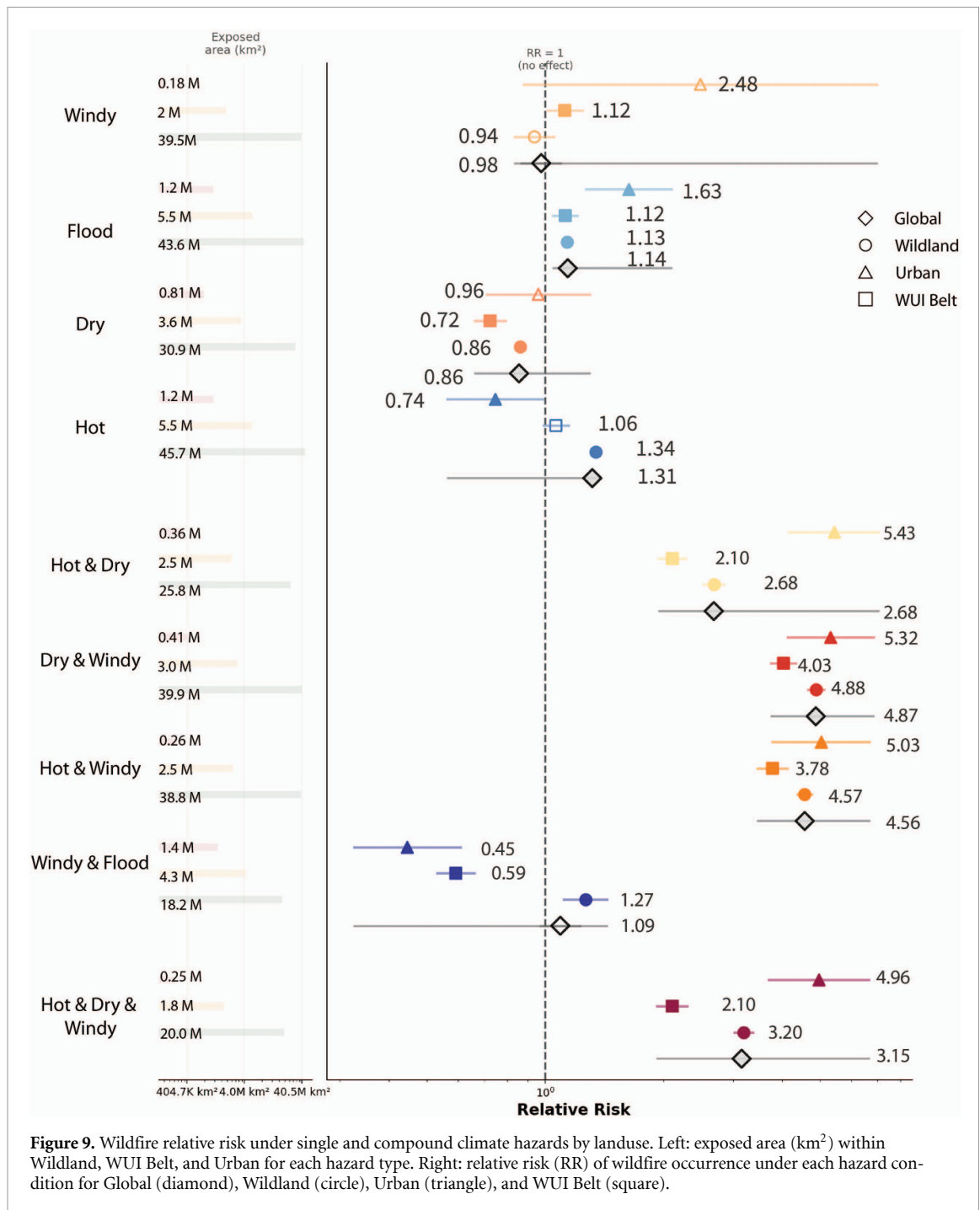
Social vulnerability and research attention show the starkest contrast of any pair. Wildland settings score 0.679 and the WUI 0.653, while urban areas score only 0.485 (figure 8(g)). These scores indicate that research concentrates where deprivation is lowest, and the least present vulnerable communities face the greatest fire exposure. When all three dimensions are combined (figure 8(h)), the WUI emerges as the most misaligned setting (0.609), exceeding both wildland (0.601) and urban (0.487) scores. Risk peaks here not because fire is most frequent, but because fire frequency intersects with concentrated vulnerability and minimal scientific attention.

The setting-stratified RR values in figure 9 have a direct methodological consequence. Standard fire weather indices, such as the fire weather index (FWI) and Keetch–Byram Drought Index (KBDI), were designed for wildland fuels and weather [68], conditions that produce the modest single-extreme response shown for the wildland category. They do not capture the urban heat island, building-modified wind fields, or infrastructure-mediated ignition that

produce the much steeper compound response in the urban category [62]. Single-variable thresholds therefore systematically underestimate risk in urban and WUI environments, making setting-adapted compound indices an urgent methodological need. The WUI belt remains the critical blind spot: in figure 8(h) it combines the highest physical risk, greatest social vulnerability, and least research coverage of any setting.

6.2. Propagation pathways of wildfire-linked impacts through urban systems

Wildfire-linked disruption propagates through urban systems long after flames are extinguished. Figure 10 maps this propagation across eight stages from pre-fire conditions to socioeconomic outcomes. The left half reflects actual literature counts from 2368 studies (the subset of the corpus with any cascade-stage signal); the right half represents the expected urban cascade chain structured according to IPCC AR6 WGII frameworks [8, 17], with dashed outlines marking nodes that the framework identifies as critical but that



lack substantive evidence. This figure maps the distribution of research attention across hypothesized cascade stages and links; it does not estimate the strength, direction, or causality of cascade mechanisms. The contrast is stark: evidence thins sharply as the chain extends into urban systems. Of 851 studies that mention at least one urban-system layer, approximately 80% address only a single stage. Fewer than 7% connect two or more stages, and fewer than 1% connect three or more. The wildfire-linked literature is overwhelmingly node-based rather than chain-based.

The sharpest evidence deficit lies in the Service Disruption column, visible in the figure as dashed outlines for healthcare access, water services, and

mobility. The IPCC framework places infrastructure interdependence and service loss at the center of urban cascade propagation. Yet our review data shows that fewer than 1% of wildfire-linked studies register explicit service disruption signals (outages, road closures, boil-water notices, shelter capacity limits). This gap breaks the evidence chain. On the left, hazard characterization and infrastructure damage are well studied. On the far right, long-term health outcomes, economic losses, and displacement are documented by recovery research [41, 49]. Between them, the mechanisms that translate physical damage into population-level service failure remain largely uncharacterized. Without this middle link, the field

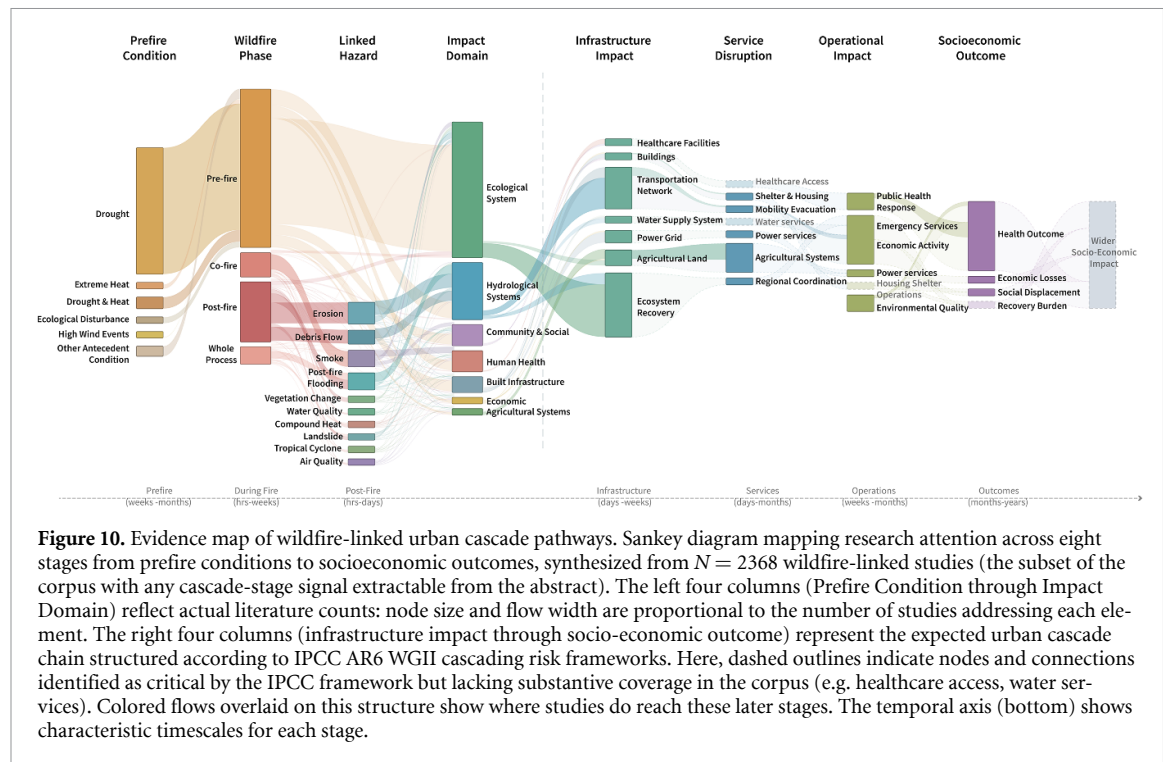


Figure 10. Evidence map of wildfire-linked urban cascade pathways. Sankey diagram mapping research attention across eight stages from prefire conditions to socioeconomic outcomes, synthesized from $N = 2368$ wildfire-linked studies (the subset of the corpus with any cascade-stage signal extractable from the abstract). The left four columns (Prefire Condition through Impact Domain) reflect actual literature counts; node size and flow width are proportional to the number of studies addressing each element. The right four columns (infrastructure impact through socio-economic outcome) represent the expected urban cascade chain structured according to IPCC AR6 WGII cascading risk frameworks. Here, dashed outlines indicate nodes and connections identified as critical by the IPCC framework but lacking substantive coverage in the corpus (e.g. healthcare access, water services). Colored flows overlaid on this structure show where studies do reach these later stages. The temporal axis (bottom) shows characteristic timescales for each stage.

cannot explain why identical infrastructure damage produces temporary inconvenience in one context and systemic breakdown in another.

This missing middle reflects a deeper structural problem: the cascade concept is frequently invoked but rarely operationalized in wildfire research. Despite IPCC's emphasis on cascading risk, fewer than 4% of wildfire-linked studies in our corpus are classified as cascading under the taxonomy developed in figure 3. The literature instead frames multi-hazard risk primarily through preconditioned, triggered, or concurrent interactions. This is not merely a labeling issue. It reflects the absence of study designs that trace propagation across system boundaries. The infrastructure interdependency literature has developed models for cascading failure in power-water-transport networks [20, 69], but almost none of this work is calibrated to wildfire events. The wildfire literature documents structure loss, evacuation, and suppression in detail. Systematic post-event analyses that follow how a specific infrastructure failure propagates through service networks to produce population-level harm are scarce [70].

The smoke pathway illustrates how these gaps manifest even where evidence is strongest. Smoke has the most robust epidemiological foundation of any wildfire-linked hazard [7, 71]. Yet this evidence concentrates on two ends of the cascade chain: pollutant characterization (left side of the figure) and health outcomes (far right). In between, the urban service chain is largely empty. Fewer than 1% of smoke-mentioning studies register service disruption signals such as school closures or healthcare system surge.

Only 2% reference air quality monitoring infrastructure. A systematic imbalance within the smoke literature worsens the problem: fine particulate matter pollution ($PM_{2.5}$, aerosol) appears in over half of smoke studies, while air quality governance terms (monitoring networks, regulatory thresholds) appear in fewer than 3%. Smoke increasingly coincides with extreme heat [72], yet the implications for urban service chains remain poorly studied (section 6.2). School closures reduce workforce participation [73]. Clean-air shelters face capacity limits during extended events [74]. These downstream consequences are widely recognized but almost never studied as part of the propagation chain. The pathway with the strongest health evidence thus has the weakest systems-level characterization.

Three reinforcing barriers explain why the evidence base remains node-based. First, disciplinary boundaries partition the cascade chain: fire ecology and atmospheric science dominate the left side while infrastructure engineering and public health anchor the right, and the middle stages fall between jurisdictions. Second, the data needed to trace service disruption (outage logs, road closures, shelter utilization) are recorded by utilities and municipalities, not by the satellite and sensor systems that fire scientists use. Third, single-hazard study designs structurally limited chain-based inference, because tracing propagation from a burned watershed through a damaged water treatment plant to a population-level service outage requires designs that span physical, engineering, and social-system boundaries.

Table 3. Strategy domains mapped to interaction types, evidence strength, and priority actions. Evidence strength reflects the proportion and quality of studies identified in the systematic review.

Domain	Primary interaction types	Evidence strength	Key barriers	Priority actions
Data & methods	All types	Moderate (wildland); Weak (urban)	<5% of studies validated for urban settings; methods from ecology poorly transfer to built environments	Hybrid statistical–process models; interpretable ML; urban-adapted compound indices
Monitoring & early warning	Preconditioned; Triggered; Concurrent	Moderate	Sparse sensor coverage at the WUI; smoke forecasts not linked to health vulnerability maps	Sensor densification in WUI; integrated smoke and heat warnings; 24 to 72 h post-fire debris flow tools
Prevention & mitigation	Preconditioned; Triggered; Cascading	Strong (fuel treatment); Weak (urban governance)	Workforce limits on prescribed fire; liability frameworks; only 2.9% of studies address governance	Scale WUI fuel treatments; enforce building codes for existing structures; watershed-to-tap risk assessment
Preparedness & response	Concurrent; Spatially compounding; Triggered	Weak	Only 2.9% of studies cover social science or governance; mutual aid fails under spatially compounding events	Compound-scenario evacuation planning; cross-agency decision-support tools; health-system surge protocols
Long-term resilience	Consecutive; Cascading; Amplified	Weak	Political resistance to managed retreat; insurance withdrawal from fire-prone regions	Strategic retreat in high-compound-risk zones; equity-centered buyout programs; parametric insurance for compound events

These gaps are intensified by the near-complete absence of research on interactions between propagation pathways. Smoke degrades visibility during evacuations from debris flow zones. Debris flows damage infrastructure whose failure initiates socioeconomic cascades. Infrastructure loss impairs healthcare systems needed to treat smoke-related illness. Yet fewer than 5% of studies examine interactions between two or more pathways, and evidence quality is inversely related to urban relevance: hazard processes are the most extensively studied but are examined primarily in wildland settings [13], while infrastructure cascades are most consequential for urban populations but have the weakest empirical base. These findings identify specific evidence gaps that the next section translates into the strategy domains and actionable milestones in table 3.

7. Strategies and pathways for wildfire-linked risk in urban systems

The preceding sections document a persistent misalignment between where wildfire evidence concentrates and where compound risks increasingly threaten populations. Bridging this gap requires coordinated advances across five domains (table 3; figure 4). For each domain we identify where evidence

is strongest, where it is weakest, and what the most actionable next steps are.

Three convergent evidence gaps emerge from the preceding analysis: research attention concentrates in wildland settings while compound exposure peaks in urban and WUI areas (figure 8); urban cascade pathways are documented at their endpoints but unmapped through service-disruption mechanisms (figure 10); and fire burden, deprivation, and research neglect converge in populations with the least adaptive capacity (figure 7). Closing these gaps requires action across the disaster management cycle to which the interaction taxonomy was mapped (figure 4, table 2).

Table 3 organizes that action into five domains. Four (monitoring and early warning, prevention and mitigation, preparedness and response, long-term resilience) follow the disaster management-cycle phases used to organize the interaction taxonomy. Data and methods is added as a cross-cutting prerequisite because fewer than 5% of studies validate methods for urban settings. Each row links the domain to the primary interaction types drawn from the taxonomy, the evidence strength inferred from corpus shares (such as the 2.9% governance rate), the structural barriers identified in our review, and the priority actions where investment would yield the largest return. The subsections that follow unpack

each domain in turn, citing prior work to support or contrast the priority actions.

7.1. Data and methods for linked-hazard research

Most studies rely on methods developed for wildland conditions. Statistical approaches dominate (50.2%), followed by process-based models (15.9%), mixed methods (11.6%), and machine learning (6.5%), yet fewer than 5% of studies validate these approaches for urban applications.

Three priorities emerge. The first is coupling statistical risk models with process-based simulations. Current approaches often treat drought-to-wildfire or wildfire-to-debris-flow sequences as independent associations [75], missing the mechanistic feedbacks through which preconditioned events escalate into triggered or cascading events. Basin-scale studies integrating wildfire, hydrology, and water supply contamination show the potential of coupled approaches [76, 77], but the core barrier is observational: coupled models require co-located fire, climate, soil, and infrastructure data that few regions outside North America and Australia can provide [78].

The second priority is wider adoption of interpretable machine learning methods. Deep learning achieves strong performance for burned area mapping and ignition modeling [79, 80], but opaque model outputs lack the interpretability that land use and emergency management decisions require. Physics-informed and explainable methods offer a path forward [81], yet none have been validated for urban fire risk. Current models are trained on wildland datasets that omit the parcel-level building and infrastructure data on which urban decisions depend.

The third is adapting compound fire weather indices to urban settings. Existing indices perform well in wildland contexts [68] but do not account for urban heat island effects, building-modified wind fields, or infrastructure-mediated cascades. The adaptation has not occurred because these indices were designed within wildland agencies, and urban meteorological, building, and infrastructure data sit in separate institutional silos. The projected increase in compound fire weather [62] makes this urgent. A realistic milestone is validated urban compound indices for at least three WUI metropolitan regions within five years.

A prerequisite is resolving how fire spreads within WUI communities. Current wildland models treat structures as non-burnable barriers [82]. Community-scale simulations show that home hardening and defensible space can reduce structure loss by up to 50%, but only at the community level [83]. Extending these models to compound conditions and infrastructure interdependencies remains unresolved.

7.2. Monitoring and early warning

Effective early warning must track not only fire ignition and spread but also antecedent conditions and downstream cascades (table 2).

For preconditioned interactions, the link between regional climate anomalies and burned area is well established [84]. Seasonal drought forecasts can provide months of lead time. What remains limited is translating these relationships into warnings relevant to urban populations. Low-cost air quality monitors, IoT-enabled weather stations, and satellite-derived fuel moisture products [85] can fill observation gaps at the WUI, but sensor deployment has concentrated in well-resourced jurisdictions. Dense monitoring in peri-urban New South Wales has improved early warning in Australia, yet equivalent networks remain absent in Southeast Asian and sub-Saharan African fire regions where the need is greater [86].

Wildfire smoke is the most spatially extensive hazard affecting urban populations, yet only 3.2% of studies address smoke-health interactions in urban settings. Operational forecasts now provide 48- to 72 h PM_{2.5} predictions with reasonable skill [87], but the bottleneck is integration into public health response. Smoke forecasts are rarely linked to vulnerability maps showing where sensitive populations, healthcare capacity, and clean-air shelters are located. Compound smoke-heat events are becoming more frequent [50], yet whether their joint health effects are synergistic or additive remains unresolved. Current systems, which address smoke and heat separately, cannot capture this interaction.

For triggered interactions, the window between fire containment and first significant rainfall is the highest-risk period for debris flows, as hydrophobic soil conditions amplify runoff from even moderate storms [88]. The USGS post-fire debris flow system translates burn severity maps into susceptibility estimates within 24–72 h [89], but it was designed for wildland watersheds. Urban drainage networks, where impervious surfaces alter runoff pathways, remain outside its scope. Extending this tool to urbanized catchments is among the highest-value monitoring investments the field could make.

7.3. Prevention and mitigation

Prevention strategies span scales from landscape-level fuel management to parcel-level building hardening. Evidence is strongest for fuel treatments, moderate for land use planning, and weakest for governance.

Among all prevention strategies reviewed, fuel treatment is a well-documented prevention success. Thinning and prescribed fire reduce fire severity and promote recovery [90], with effects persisting over multiple years [91]. Treatments at the WUI yield disproportionate protection by addressing both flame contact and ember exposure [92]. The barrier is not

efficacy but scale: expanding treatments confronts workforce shortages, smoke management concerns, and liability frameworks that discourage proactive management.

Land use planning and building codes are potentially the most cost-effective prevention tools, yet implementation evidence is thin. Structure density, arrangement, and materials strongly influence fire transmission at the WUI [93]. However, building code enforcement varies widely, and older, higher-risk structures built before current standards are often exempt from compliance. Only 2.9% of studies address governance, so it remains unclear which regulations are most effective at reducing risk or whether compliance differs across communities of varying income and resources.

Critical infrastructure both causes ignitions (e.g. utility-started fires) and propagates cascading failures when damaged. Public Safety Power Shutoffs reduce ignition risk but impose burdens on medically vulnerable populations [94]. Water systems face a different vulnerability: post-fire contamination can degrade water quality for months to years [95], and debris flows damage treatment infrastructure. Integrated watershed-to-tap risk assessment is an emerging priority, but no jurisdiction has implemented one, largely because water utilities, fire agencies, and land managers operate under separate mandates.

7.4. Preparedness and response

Emergency management for wildfire-linked hazards requires coordination across multiple agencies, jurisdictions, and hazard phases, yet governance perspectives remain scarce in the literature (table 3). This is the weakest evidence domain and the one where targeted investment would yield the greatest operational returns.

Evacuation under compound conditions creates failure modes that single-hazard planning does not anticipate. When fire coincides with extreme heat, outdoor evacuation becomes hazardous for populations lacking air-conditioned transport [72, 96]. When post-fire rainfall threatens debris flows, evacuation routes may already be compromised [88, 93]. Models such as WUIVAC demonstrate how evacuation triggers can be integrated with wildfire spread projections [97], but validation remains limited to a small number of scenarios. More broadly, evacuation compliance depends on risk perception, trust, and practical constraints that vary across communities and have received far less attention than the physical hazards themselves [98]. The result is that evacuation plans are optimized for fire physics but poorly calibrated to human behavior under compound stress.

As shown in table 2, spatially compounding conditions cause mutual aid networks to fail. Decision-support tools that account for simultaneous fires, smoke effects on aviation, and post-fire debris flow

threats would be high-value applications. However, existing research prototypes are designed for post-hoc analysis, not for operational use under severe time pressure. The gap between research tools and operational requirements remains wide.

Health system preparedness requires surge capacity for respiratory and cardiovascular admissions alongside messaging that reaches vulnerable populations. Smoke exposure drives increases in emergency visits, hospital admissions, and mortality during major fire events [51, 99]. Joint smoke-heat effects on hospitalizations further underscore the need for compound-aware preparedness [50]. Yet health system integration into fire emergency frameworks remains incomplete. Coordination between air quality agencies, public health departments, and emergency management varies widely across jurisdictions [100]. Where integration has been attempted, results are mixed. California's interagency smoke response coordination has improved alert issuance but has not demonstrably reduced hospitalizations during major events. The most achievable near-term milestone is standardized protocols for compound smoke-heat events that specify trigger thresholds, agency responsibilities, and communication templates.

More broadly, translating the interaction taxonomy into operational guidance requires practitioner-facing decision trees that link observed conditions to recommended response protocols. These should be tested through cross-agency coordination pilots for compound fire events, with explicit protocols specifying who acts, when, and with what authority. History-informed hazard planning offers a tested model. In Montecito, analysis of historical records and regulatory maps revealed that exposure in high-hazard zones nearly doubled over 50 years despite policies discouraging development, and that conventional flood insurance maps failed to capture post-fire debris flow risk [101]. Adapting such approaches to compound wildfire-linked hazards is a feasible next step.

7.5. Long-term resilience and transformation

As fire seasons lengthen and compound dry-hot-windy extremes increase [88], restoring pre-disaster conditions after each event is no longer a sufficient strategy. Managed retreat is entering policy discussions after repeated losses [102]. Buyout programs, transfer of development rights, and updated zoning offer practical mechanisms, but implementation must navigate property rights, community attachment, and fiscal constraints. In practice, outcomes have been uneven: federal buyout programs have disproportionately served wealthier communities, and processing times often exceed five years [103]. Incorporating compound risk metrics into buyout eligibility criteria, replacing single-hazard return periods, is a concrete next step.

These patterns reflect deeper structural inequities. The vulnerability-research mismatch in figure 7(d) shows that wildfire research is most sparse precisely where social deprivation is highest, a pattern documented in low-income WUI housing, evacuation barriers, and slower recovery outcomes [67, 86]. Procedural vulnerability frames why: historical policy decisions in zoning, infrastructure investment, and recovery design concentrate risk in marginalized populations [104]. Equity-centered adaptation requires both redirecting resources toward underserved communities and restructuring decisions so affected populations help set priorities. An immediate step is requiring vulnerability assessment as a standard component of wildfire research funding proposals.

Insurance market retreat from fire-prone regions reflects a direct consequence of this evidence gap: risk models built on single-hazard, wildland-centric data cannot price compound urban exposure accurately [105]. Developing compound-aware actuarial frameworks is therefore both a financial and a scientific priority.

A cross-cutting barrier limits progress across all five domains: the knowledge system is structured around post-event response cycles rather than long-term resilience. The deficit is most acute at the WUI, where no region maintains interoperable databases linking fire occurrence, building stock, infrastructure condition, and social vulnerability at the resolution that compound risk models require. Three shifts should be prioritized. First, standardized WUI-scale data collection programs, co-designed with local agencies, would provide the inputs needed to calibrate and validate urban fire risk models. Second, open repositories for post-fire impact assessments would enable cumulative learning across events. Third, these platforms require dedicated long-term funding, analogous to long-term ecological research networks, because the agencies involved operate on incompatible timescales, none matching the decades-long horizon over which fire regimes, land use, and climate interact.

8. Constraints and limitations

The LLM-assisted pipeline enabled consistent classification across 3625 records, a volume at which manual coding would sacrifice either coverage or reproducibility. Because the pipeline operates on abstracts, studies that address equity or cascade mechanisms only in their main body may be undercounted. Validation against 350 high-ambiguity records yielded mismatch rates between 0.3% and 4.9% (table 1), placing agreement at or above 95% under the most demanding test conditions. Research attention is allocated to grid cells in proportion to each study's geoparsed footprint (section 3.2.5).

The precision of this surface depends on the precision of the bounding boxes extracted by the LLM pipeline; approximately 8% of studies returned no geoparseable region and were excluded from the analysis. Bounding boxes are rectangular approximations that may include cells outside irregular study areas. Around 60% of contributing studies have country-scale or coarser footprints. Therefore, cell-level scores should be read as approximate, while SREX-region-level inference averages across many cells per region and is robust to grid resolution (supplementary material C5.9).

Fire occurrence and climate hazard frequencies were aggregated to annual totals across 2001 to 2024. This averaging captures climatological associations but does not resolve the seasonal and short-term dynamics that drive individual events. RR estimates therefore reflect mean annual co-occurrence rather than event-scale forcing. MODIS also underdetects small and low-intensity fires prevalent in tropical regions [13], likely underestimating fire burden in the Global South. Because this bias suppresses fire counts where convergent deficits are most severe, the mismatch indices reported here likely underestimate the hemispheric research gap.

The mismatch framework locates structural co-occurrence of fire burden, research neglect, and social vulnerability, but the GRDI does not capture institutional capacity, governance quality, or community-level adaptive resources. Aggregating the 10 m WUI map to a 10 km grid smooths fine-scale heterogeneity, potentially masking localized pockets of high exposure. These omissions affect the precision of individual grid cell estimates but do not alter the overarching geographic pattern. The hemispheric concentration of unmet research need in the Global South persists across every analytical variation reported in this review.

9. Conclusion and outlook

What makes wildfire dangerous to cities is often less the fire itself than what it sets in motion. The most consequential impacts propagate through urban networks long after flames are contained: smoke degrades regional air quality, burned slopes produce debris flows, utility outages trigger service cascades, and health burdens accumulate unevenly across neighborhoods. However, the research record remains concentrated in wildland settings. In our corpus of 3625 studies (2000–2025), 96.0% focus on wildland contexts; only 0.9% examine WUI environments and 3.1% urban areas, even though these settings account for over 33% of global fire occurrence.

Three findings from this review characterize this gap and motivate a path forward. A decision-tree interaction taxonomy reconciles inconsistent terminology across disciplines and maps interaction types

to management phases (figure 3). Global mismatch mapping identifies where deprivation, fire burden, and research neglect coincide (figure 7). Setting-stratified compound analyses show that urban and WUI fire risk responds to co-occurring extremes in ways that single hazard frameworks do not capture (figure 9). Together, these findings call for a reorientation: wildfire science should treat WUI and urban fire as a coupled hazard–governance problem, not as a boundary condition of wildland research. Closing these gaps requires coordinated progress across five priority areas:

Rebalancing geographic coverage is the most immediate priority. The triple mismatch framework identifies where investment is most urgent, but structural barriers impede reallocation. Fire-prone regions in the Global South lack the monitoring infrastructure and institutional capacity that underpin risk assessment elsewhere. Within the near future, the field should establish satellite data-sharing agreements between global fire monitoring programs and national agencies in at least five high-mismatch regions identified in figure 7. Collaborative partnerships that pair Global South institutions with established fire science groups, structured to build local capacity rather than extract data, would begin to close this gap.

Compound hazard assessment for urban conditions has lagged far behind. The nonlinear risk amplification documented in urban settings (figure 9) requires indices that capture co-occurring extremes rather than single variables. A tractable starting point is validating urban compound fire weather indices for at least three major WUI metropolitan regions, incorporating heat island effects and building-scale wind data. The field also needs open validation benchmarks for urban fire risk models, analogous to those standard in weather forecasting.

Integrating equity from project design through implementation is the standard the field has not yet met. The inverse relationship between deprivation and research attention (figure 7) means that adaptation guidance is least informed by equity considerations where equity stakes are highest. Beyond procedural reform, the field needs vulnerability frameworks calibrated to Global South conditions, where adaptive capacity constraints differ qualitatively from those in wealthy nations.

Converting the interaction taxonomy into practitioner protocols is the most direct route from classification to management action. The taxonomy developed in this review maps interaction types to management phases (table 2), but translation into operational protocols remains incomplete. The most actionable next step is producing practitioner-facing decision trees that link observed conditions to recommended response protocols, tested through cross-agency coordination pilots in at least two jurisdictions.

Building WUI-scale data infrastructure for long-term resilience addresses a cross-cutting barrier that limits progress across all four preceding directions. The knowledge system is structured around post-event response cycles rather than sustained risk reduction. WUI environments, where compound risk peaks and research coverage is weakest, lack the foundational data needed to build, calibrate, and validate urban fire risk models. Closing this gap requires three investments. First, systematic WUI-scale data collection that links fire occurrence, building attributes, infrastructure condition, and social vulnerability in interoperable formats. Second, open post-fire impact repositories with standardized reporting templates that enable cumulative learning across events and regions. Third, dedicated long-term funding for maintaining these platforms, analogous to long-term ecological research networks, so that the knowledge base evolves with shifting fire regimes rather than resetting after each disaster cycle.

In a warming world, wildfire-linked risks will continue to intensify and migrate toward populated areas. Wildfire science has documented these gaps individually. This review shows where they converge and which areas bear the greatest cost of their persistence. The quantitative framework introduced here makes that convergence legible and actionable. It identifies specific regions, settings, and research directions where investment would address the largest measured misalignments between fire risk, social vulnerability, and scientific knowledge. The steps outlined above are ambitious but tractable. Achieving them would shift wildfire science from documenting past patterns to anticipating the compound, cascading, and inequitable risks that will define urban fire management in the coming decades.

Acknowledgments

The authors acknowledge support from the Graduate Research Fellowship at the UC Berkeley College of Environmental Design, funded by a gift from Autodesk, and thank the anonymous reviewers for their constructive and thoughtful feedback.

Funding

This research received no external funding.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/xinyizh-sydney/Wildfire-Linked-Hazard-Literature-Review> [106]. The global spatial analysis uses four publicly available datasets: MODIS

burned-area product MCD64A1 Collection 6.1 (NASA LP DAAC, <https://lpdaac.usgs.gov>); ERA5-Land daily aggregates (Copernicus Climate Data Store via Google Earth Engine, <https://developers.google.com/earth-engine/datasets>); global 10 m wildland–urban interface classification [6]; and the Global Gridded Relative Deprivation Index (NASA SEDAC, <https://sedac.ciesin.columbia.edu>). The literature corpus was retrieved from the Web of Science Core Collection, which requires institutional subscription; full search query strings are provided in supplementary material A. LLM classification prompts, decision rules, and codebook definitions are documented in supplementary material B2.

Methods and Extended Results available at <https://doi.org/10.1088/1748-9326/ae8039/data1>.

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